Predicting Artist Success using Social Media Data

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June 29, 2021

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The A&R departments at music labels are in charge of scouting and developing musical artists. These groups have historically relied heavily on "gut feelings" to make decisions about which artists to sign. This is largely driven by the industry wide "Nobody-Knows-Anything" mantra attributed to Hollywood producer William Goldman, describing the apparent impossibility of defining what content will be successful due to the high variance in results. In this research, we study how audience data extracted from social media can play a central role in informing decision makers and help mitigate the risk and uncertainty inherent to the artist scouting process. We employ machine learning and statistical methods to predict two different measures of success: whether the artist signs a deal to produce an album with a major record label and whether a new artist's work will appear in a Billboard chart. We train and test our models using data from 188,734 videos on YouTube, representing 6,259 artists that released music in the US in the period of 2017-2019.

Our classification results show that audience metrics from YouTube can be used to predict these two major milestones in an artist's career with some level of success. This finding contributes to the understanding of industry managers looking to increase the use of data in their business decisions, and the broader academic literature analyzing how social media data can be used to predict practical business outcomes.

Key words: online platforms, predicting success, social media data

1. Introduction

Record labels have historically relied on "gut feeling" to make decisions on which acts to sign. These decisions are usually made by the A&R (Artist and Repertoire) departments, the organization within a music label responsible for scouting and signing artists (IFPI 2019). This is largely driven by the industry-wide "Nobody-Knows-Anything" mantra attributed to Hollywood producer William Goldman, describing the apparent impossibility of defining what content will be successful due to the high variance in results (Goldman 1983). However, as discussed by Smith and Telang (2016) and Hennig-Thurau and Houston (2018), the emergence of very successful, heavily data focused firms show that technological advancements and data analytics can play a central role in informing entertainment industry decision-makers to help mitigate the risk and uncertainty inherent to the processes of the entertainment markets, among those the artist scouting process.

Record companies invest more than one-third of their global revenues ($5.8 billion in 2019) in A&R and marketing each year to break, develop and support artists (IFPI 2019). Furthermore,

historically only 20 percent of the recordings released by labels recoup their costs (Rivkin 2000). Alongside this highly risky business, the rise of social media platforms and streaming services has led to a large increase of musical content being posted online. This means that A&R department are faced with the challenging task to navigate an enormous and increasing pool of content to decide what acts are worth signing and developing. Motivated by this, we are interested in understanding the relationship between digital platform metrics and the success of an artist in an early stage in their career, particularly if observable social media metrics early in an artist's career can be a reliable indicator of a successful career.

Predicting future career trajectory using measurable productivity outcomes has been of interest to both academic researchers and firms. Research on "Science of Science" uses large-scale data on the production of science to search for patterns describing the structure and evolution of science and early scientific careers (Clauset et al. 2017, Fortunato et al. 2018). Most closely related to this research is Arakelyan et al. (2018) who use concert and venue data of previous performances to predict whether an artist will be signed by a major record label, and if an artist will play at a particular venue.

Other researchers have studied how data from digital platforms can be used to forecast if an item will become popular. Kim et al. (2014) find that music-related hashtags from Twitter are highly correlated with general music trends and can be used to forecast Billboard rankings and hit songs. Soares Araujo et al. (2019) use a song's "popularity" score computed by Spotify, in addition to acoustic features of the song to predict if it will appear on the Spotify's Top 50 Global ranking. They show an AUC of 80 percent when predicting the appearance of the song in the chart two months in advance.

Other studies have focused on understanding what drives the popularity of a digital item. Shulman et al. (2016) how the popularity of songs, movies and tweets can be predicted using data describing the speed to acquiring early adopters, while the properties of the item (such as the structure of the network where it is shared and the content of the item itself) do not contribute much to the prediction. Similarly, Szabo and Huberman (2008) show that the number of views a YouTube video receives a few days after posted is strongly correlated with the views at a later time, and thus can be used as an effective predictor in a popularity classification task.

Efforts have also been made to develop tools that serve as decision support for decision makers in the entertainment industry. Eliashberg et al. (2000) developed MOVIEMOD, an interactive Markov Chain model designed to generate box-office forecasts and to support marketing decisions for a new movie after it's production but before it's release, by collecting survey data from potential

viewers. Similarly, a handful of companies have emerged that aggregate music data and display dashboards and useful data points to music executives, such as Chartmetric[1] and Sodatone[2].

The field of Music Information Retrieval (MIR) has been concerned with retrieving and analysing multifaceted information from large music databases (Herremans et al. 2014, Downie 2003). Most of the research in this field has been focused on automatic classification of songs based on a diverse set of acoustic features and artist characteristics, but there has also been interest in MIR systems for hit prediction. The Echo Nest, a music intelligence company acquired by Spotify in 2014[3] and the Million Song Dataset (Bertin-Mahieux et al. 2011) have been widely used by researchers for these tasks. Herremans et al. (2014) builds a classifier top predict "Top 10" dance hits, using data from dance songs extracted from the Echo Nest, including basic musical features and temporal features. Their logistic regression model shows decent performance, capturing almost 84 percent of hits while maintaining a False Positive Rate of 32 percent.

In this study, we explore the extent to which social media data can be used as a signal to detect artists who have the potential of becoming successful. Leveraging a unique data set of YouTube engagement metrics, musical releases and popularity charts, we attempt to predict whether early career musicians will reach two major milestones: working with a major record label and appearing in a Billboard chart for the first time. Our results show that features derived from YouTube videos describing an artist's average performance and growth in the platform are predictive of our two metrics of success, but are not sufficient to accurately capture a majority of potentially successful artists. Specifically, we achieve an Average Precision of 9 percent for predicting whether an artists gets to work with a major record label, and 13 percent for predicting their first Billboard appearance. In particular, a Top 50 ranking of artists created using our models' predicted probability of success contains 16 percent of artists that eventually sign with a major label, and 19 percent of artists that end up appearing on a Billboard chart. These results suggest that YouTube performance is more predictive of an artist breaking into popularity charts than it is of an artist signing with a major record label. Our results motivate the further study of the association between observable social media performance and more objective and readily available measures of success.

1.1. Predicting Success of an Early Career Artist

Unlike the pre-Internet years, where scouts often visited venues to discover new talent, the industry is increasingly looking to rely on social media to discover new talent. This shift is driven by the

[1] `https://www.chartmetric.com/`

[2] `https://www.sodatone.com/`

[3] https://www.theguardian.com/technology/2014/mar/06/spotify-echo-nest-streaming-music-deal

explosive growth of social media and streaming services. As stated by Spotify[4], close to 60,000 thousand tracks are uploaded to the platform every day, a 50 percent increase from the 40,000 tracks per day reported in 2019[5]. At this pace, Spotify will be home to nearly 90 million songs by the end of 2021. A similar story can be told for YouTube[6], where Music represent 5 percent of the content and accounts for 20 percent of the views, and newer platforms such as TikTok[7], where songs driven by viral trends top the popularity charts. Content creation, and particularly music creation, is part of this growth. A drop in the costs of producing, and the ability to reach worldwide audiences through social media and digital platforms, has made it easy for aspiring artists to post their content online and create a following. Our goal is to leverage the wide availability of this online content to search for and study patterns that characterize an artist's early career and can be predictive of future success. Accurately predicting future popularity can greatly reduce the time, effort, and cost incurred by A&R scouts searching for acts worth pursuing.

Success can be measured in multiple ways, and can be defined differently according to the stakeholder's goals and the stage of an artist's career. One can define success based on how well an artist's music is received on the market (as measured by the number of plays of songs and albums across platforms), sales (coming from concert tickets as well as digital and physical inventory), or by reaching particular career milestones (signing with a major record label or playing at a famous venue). In this research, we will focus in two metrics of success for artists early in their careers, particularly for those who have not yet signed with a major record label.

The first metric is the first time an artist releases an album with one of the big three major record labels (Sony Music Group, Universal Music Group, and Warner Music Group) or their subsidiaries. This is a major milestone in an artist's career, given that the major labels are able to provide artists with a myriad of resources to help them achieve success. These range from global distribution of their music, marketing, press and publicity; access to creative partnerships for collaborations in production and songwriting; and even creative teams to help them develop their individual and visual identity for album artwork and music videos (IFPI 2019).

The second metric of success describes the first time that an artist breaks into one of Billboard's top music charts[8]. Billboard publishes charts based on the popularity of an artist's music on multiple channels[9]: sales charts, capturing top performing artists in terms of digital and physical

[4] https://www.musicbusinessworldwide.com/over-60000-tracks-are-now-uploaded-to-spotify-daily-thats-nearly-one-per-second/

[5] https://www.musicbusinessworldwide.com/nearly-40000-tracks-are-now-being-added-to-spotify-every-single-day/

[6] https://pex.com/blog/what-content-dominates-youtube/

[7] https://www.rollingstone.com/pro/features/tiktok-video-app-growth-867587/

[8] https://www.billboard.com/

[9] https://www.billboard.com/p/billboard-charts-legend

sales (covering 90 percent of the US music retail market); airplay charts, compiling information from radio stations from 140 markets across the US; and streaming charts, which rank music based on their performance on the main on-demand and online radio platforms. Billboard also produces charts summarizing artists performance across the three channels. Appearing on a chart is therefore a powerful indication of an artists popularity, and a learning model that identifies artists who are likely to appear on these charts can reduce uncertainty in the scouting process, thereby reducing effort and time spent browsing content and increasing the overall effectiveness of signing decisions within the label.

2. Data

The data used in this project comes from three sources: video metrics data from YouTube, metadata for artists from the online music database Discogs, and chart data from Billboard[10].

Our social media data consist of metrics from YouTube videos posted on the site from 2017 to 2019. These data were collected and provided by PEX[11], a company focused on helping media companies identify, attribute, and license video content in real time. PEX scrapes YouTube for videos as soon as they are posted, and captures publicly available metadata for the video at the time of collection. This process allows us to have a history of video metrics at different points in time for the artists and songs in our study.

To reduce the universe of videos and users, we focus on YouTube videos with the user generated tag "music." This results in 170M videos posted by 19M users during our period of study. For every video, we are able to observe the number of views, likes, dislikes, comments and shares for 1 or more points in time after their creation date. Due to resource limitations, and the massive amount of content published in YouTube, PEX's data collection is not time exhaustive. For each video we have different number of observations, distributed over the time range listed above. In Figure 1, we show an example of a set of videos corresponding to singer Christine Lalama, who signed for Atlantic Records in 2018[12]. Each line in the plot represents a time series of cumulative views for a video.

Discogs[14] is an online crowd-sourced music database that contains information about music recordings, artists and labels—including information about the relationship between these entities. For each recording, we can identify, among other things, the artist (or artists) that released it, the record labels involved in the release, genre tags and the date of release. The database is composed

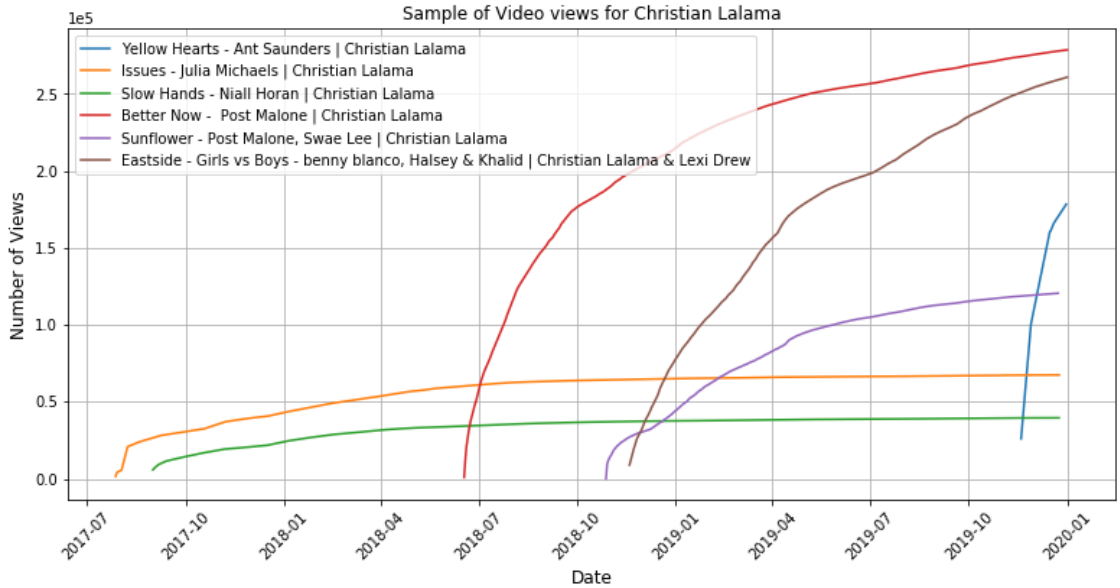
[10] `https://www.billboard.com/charts`

[11] `https://www.pex.com`

[12] `https://www.christianlalama.com/bio/`

[14] `https://www.discogs.com`

Figure 1 **Example of time series data for singer Christian Lalama[13]. Each line shows the views trend for a particular YouTube video. For this example, we have have complete observations of cumulative daily views through early 2020.**



of worldwide recordings for all genres, and by the time of data collection in July 2020, it contained entries for over 7M artists, 1.5M labels and 12.5M releases. We used this data to identify a set of musical artists, and to create our first metric of success, which include if and when an artist released their first album with a major record label.

To create our second measure of success, indicating if and when an artist an artist appeared for the first time in a Billboard chart, we collected a set of 151 charts using the Python Billboard API developed by Guo (2021). These data correspond to the period from 2005-2021, or the first date available after 2005 for some of the newer charts. Each chart provides information on the artist name, their song or album, the rank in the chart and the date of publication. The complete list of the charts collected is included in the Appendix in A16.

2.1. Data Set Creation

Given the large number of videos in the YouTube data set, and the fact that any user can upload a piece of content and tag it as music, regardless of them being in fact professional musicians, we use the Discogs data set as our main source to identify artists. Discogs has a unique ID associated to each artist, as well as their history of recordings and their collaborations with different record labels for each release. Given the time frame of the YouTube data, we identified and collected artists in Discogs that have released any recording after 2018 through 2019. To be included in our data set, artists should have released music in the US, and should not have an album with a major record label before 2018. Following this procedure, we identify a total of 42,560 artists.

There is not a universal ID or data set that allows us to link an artist's YouTube channel to an artist profile on Discogs. Therefore, we relied on artists' name to match artists across our different data sources. Exact name matching alone misses a considerable number of potential pairings, since YouTube channel names usually include additional words like "Music" or differ in spacing and abbreviations. Therefore, we employ a fuzz matching technique based on computing the TF-IDF statistic, first introduced by Salton and Buckley (1988). TF-ID stands for Term Frequency – Inverse Document Frequency and is computed as follows:

$$\text{TF-IDF}(t) = TF(t) * IDF(t)$$

$$TF(t) = \frac{\text{Number of times term t appears in an artist Name}}{\text{Total number of terms in the artist name}}$$

$$IDF(t) = \log\left(\frac{\text{Total number of artist names}}{\text{Number of artist name with term t in it}}\right)$$

After computing the TF-IDF for each term of the artist's name, we compute the cosine distance between the vectors of the two lists of names, resulting in a score between 0 and 1 for each pair of names. For each Discogs's artist name in our list, we select the YouTube channel name that has the highest score, filtering only for pairs of names that have a score of 0.9 or higher. Following this process, we were able to match 35,791 (84 percent) of the Discogs artists with a YouTube channel. The same procedure was followed to match artists across all our different data sources.

3. YouTube Video Features

Streaming music has been the top source of music revenue since 2017, with 62.1 percent of the total share as of 2020 (IFPI 2021). Among the streaming platforms, YouTube is the most popular with 44 percent of adults in America saying they use YouTube to listen to music in 2020[15]. Even though YouTube is not exclusively a music service, music represents 20 percent of their total views in 2018, and close to 83 percent of the videos with more than 1 billion views in 2019 fall into this category[16]. Thereby, It is not surprising that well-established artists publish content that amasses large amounts of views and followers on such platforms.

We hypothesize that YouTube video trends of a newer, less well-known artist are also correlated with their popularity and can be indicative of the artist's ability to create content that engages

[15] https://yougov.co.uk/topics/media/articles-reports/2021/03/18/services-used-stream-music-poll

[16] https://pex.com/blog/state-of-youtube-2019-music-more-valuable/

users and creates a following. We can measure the audience's engagement on the artist's music based on YouTube video metrics such as views, likes, dislikes and number of comments. Given that YouTube is one of the main platforms for artists to display their content and for users to stream and discover music, we expect that artists that show above average performance and growth in these metrics will have a higher likelihood of achieving success as opposed to artists with below average metrics.

Previous studies that attempt to predict popularity of an item use features that can be categorized in four main groups based on what aspect they describe: content, structural, early adopters and temporal (Cheng et al. 2014, Shulman et al. 2016). Content features refer to inherent characteristics of the piece of content, such as song's audio features and lyrics. Research using this type of features (Pachet and Sony 2012) has failed to found significant statistical correlations with popularity. Structural features describe the underlying social network of the producer of the content, as in the fans and collaborators of an artist that publishes a song, or the network structure of followers of a Twitter user (Shulman et al. 2016). Temporal and early adopter features capture the trends of adoption of an item, describing the amount of adopters (retweets in the case of tweets, plays and streams in the case of songs) and how fast it reaches certain threshold of early adoptions, and has proven to be strongly correlated of popularity (Szabo and Huberman 2008, Shulman et al. 2016).

In this research, we focused on features summarizing audience response at different points in time, features that are more closely related to the temporal and the early adopters categories. In the next sections, we describe the sets of features created that were used as input for the classification tasks, with the intent to predict the two metrics of success: releasing an album with a major record label and appearing on a billboard chart for the first time. Through out the paper, we'll refer to the dependant variable corresponding to the first metric of success as $Signed_i$ and $Chart_i$ for the second one.

3.1. Average Audience Engagement Features

Features that fall under this category are designed to measure the average performance of the artist's videos, reflecting the audience's engagement and interest towards their content. Audience engagement is a critical indicator that A&R takes into consideration when evaluating an artist's potential, they want to understand if an artist can connect with their audience (Rocchi 2020). We summarized an artist YouTube performance using 4 metrics: number of views, likes, dislikes and comments. These metrics are highly correlated; we observe that videos with large number of views also have a larger number of likes, dislikes and comments. Given that we have varying sampling frequency for the metrics (i.e., video metrics are sampled at different points in time that are not

Table 1 Summary statistics of daily metrics for YouTube videos

	Mean	Std. Dev.	Median	Min	Max	Number of Videos
Daily Views	450.82	23,139.48	5.27	0.00	6,786,210.01	188,734.00
Daily Comments	1.05	38.65	0.01	0.00	9,612.14	188,734.00
Daily Likes	10.90	425.13	0.19	0.00	124,095.43	188,734.00
Daily Dislikes	0.34	17.14	0.00	0.00	4,157.86	188,734.00
Days Since Created	76.75	87.20	32.00	1.00	365.00	188,734.00

equal across videos), we normalize by computing the average daily performance for each video, dividing the observed metric by the number of days since the video was posted. In Table 1, we present summary statistics for these features.

Given that an artist can have more than one video in our observation period, we compute the average metric across videos for each artist. We describe the process below:

- **Average of daily views**. Computed by taking the daily average number of views for each video for the period between creation date and the latest available update date, and then taking an average for all videos for $artist_i$.
- **Average of daily comments**. Computed following the same procedure as for views, but for number of comments.
- **Average of daily likes**. Computed following the same procedure as for views, but for number of likes.
- **Average of daily dislikes**. Computed following the same procedure as for views, but for number of dislikes.

In addition to the average, we also compute the median of the daily metrics and include the maximum for each of the metrics listed above, given that artists with unusually high performance on a single video may catch the attention of talent scouts (Rocchi 2020).

3.2. Video Performance Percentage Change Features

Features on this category are intended to capture the artist's evolution on the platform. A very important measure that labels consider when signing a new act is the artist's ability to keep and substantially grow their audience. We hypothesize that an artist's success will be correlated with particular positive growth patterns. We will capture this growth by observing the percentage change between an artist's first video and their last observed video. The features are created as follow:

- **Percentage Change in Views:** Computed by taking the percentage change between the first observed video's average daily views and the last observed video's average daily views for $artist_i$. We normalize this metric by dividing by the number of days between the posting of the first and last video observed.

- **Percentage Change in Comments:** Computed following the same procedure as for views, but for average daily comments.
- **Percentage Change in Likes:** Computed following the same procedure as for views, but for average daily like.
- **Percentage Change in Dislikes:** Computed following the same procedure as for views, but for average daily Dislikes.

3.3. Video Cluster Features

For features in this group, we are interested in identifying trends in the videos' view-count curves that are characteristic of artists that succeed against those who don't succeed. Research studying view-count dynamics in YouTube (Richier et al. 2014), identified that among a random sample of YouTube videos, 75 percent of videos can be categorized within two broad categories based on their viewership growth: non-viral content, characterized by concave shaped view-count curves; and viral content, characterized by view-count curves that first adopt a viral behavior (in a S-shaped phase) and then grow linearly.

We will capture distinctive view-count trends by creating clusters of videos based on the shape of their cumulative views, and then computing the percentage of videos that an $artist_i$ has in each of the clusters. In a study of early prediction of popularity trends (Figueiredo et al. 2016), the authors developed TrendLearner, an algorithm that accurately predicts the popularity of user generated content at a future date (measured by number of views) by assigning videos to clusters based on the shape of viewership time series, and use this as an input to regression models. Motivated by this, we hypothesize that the patterns in view-count curves may also serve as useful features for predicting an artist's future success.

3.3.1. Imputation of missing observations The clustering algorithm requires us to have equally sized time series for each video. However a majority of the videos in our data set have gaps between observations. To overcome this issue, we rely on interpolation techniques. The selection of the interpolation method did not affect our classification results significantly, so we chose piece-wise linear interpolation as our preferred method. We tested standard piece-wise linear, quadratic and cubic interpolation. Given that we observe cumulative views, we also tested cubic monotonic interpolation (Virtanen et al. 2020, Fritsch and Butland 1984). Classification and clustering results with alternative interpolation methods are shown in the Appendix.

We clustered the YouTube videos using the Dynamic Time Warping similarity measure and the k-means algorithm (Sakoe and Chiba 1978, Petitjean et al. 2011), employing the Python package Tslearn developed by Tavenard et al. (2020). Dynamic time warping is a standard technique to measure similarity between time series that vary in speed, time, or length.

Figure 2 **Resulting Video Clusters from applying the DTW algorithm to time series of the cumulative views observed for the first 60 days since video creation.**

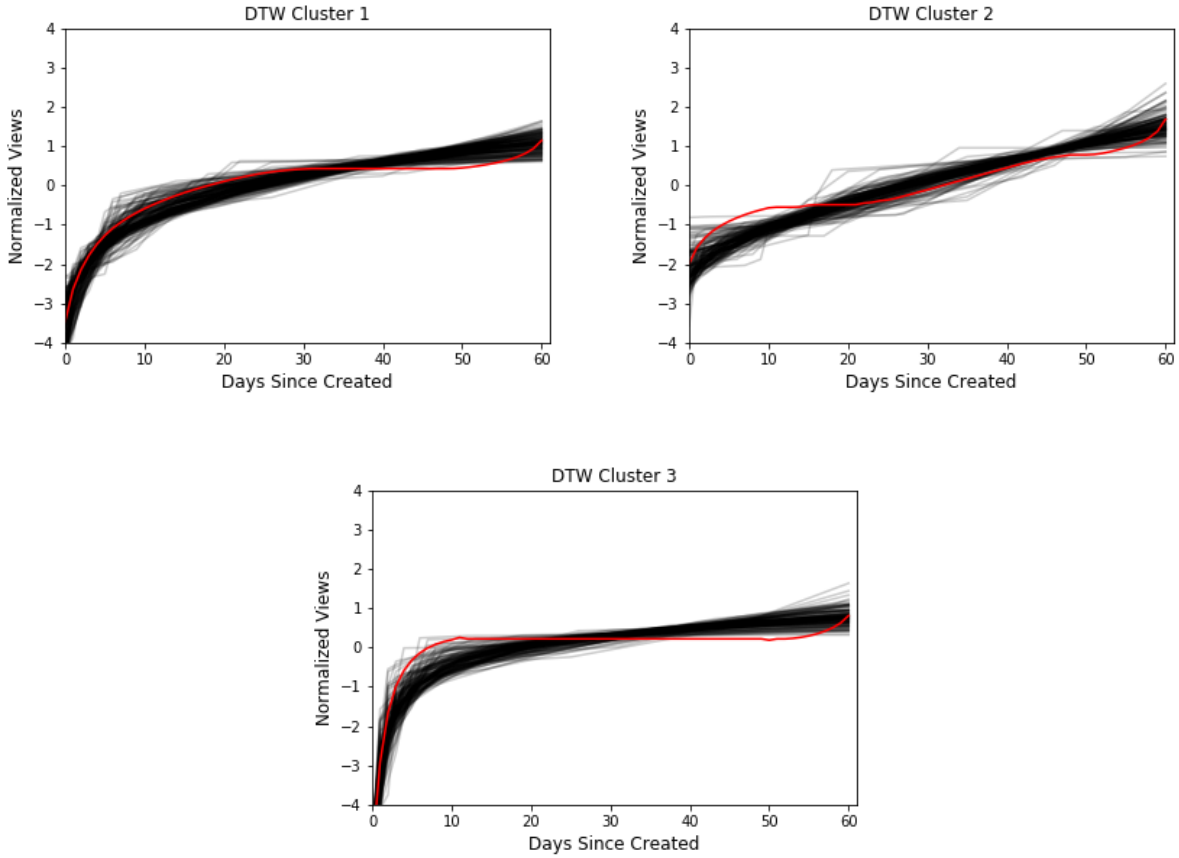


Table 2 **Summary Statistics describing Mean and average Views for Video Clusters**

Cluster	Mean Views	Std. Dev.
Cluster 1	21,685	182,343
Cluster 2	116,518	3,664,816
Cluster 3	8,929	70,823

Following Figueiredo et al. (2016), we set the number of clusters to three, and use a standard normalization for the time series. The resulting clusters can be seen on Figure 2. We notice certain patterns for each of the clusters. Cluster 2 shows an upward trend in cumulative number of views with a non decreasing rate during the first 60 days of video creation. Cluster 1 and 3 show more concave shapes, with Cluster 3 leveling off faster than Cluster 1. In average, videos in Cluster 3 reach 80 percent of their total 60-day views by day 28 after creation, while videos in Cluster 1 do so by day 11. In contrast, videos in Cluster 2 reach this mark by day 45.

As seen in Table 2, Cluster 2 has a higher average number of views by day 60 after creation date as compared to the other clusters, but has a significantly higher variance. Cluster 3 has the smallest average number of views and seems to gain most of them early after its creation date, with

Table 3 Time to J views Summary Statistics

	Mean	Std. Dev.	Median	Min	Max	Number of Videos
Time to 100 views	2.30	6.45	1.00	0.00	60.00	41,822.00
Time to 280 views	2.70	6.19	1.00	0.00	60.00	39,630.00
Time to 460 views	3.18	6.40	1.00	0.00	60.00	38,615.00
Time to 640 views	3.77	6.94	1.00	0.00	60.00	38,017.00
Time to 820 views	4.47	7.61	2.00	0.00	60.00	37,573.00
Time to 1000 views	5.43	8.69	2.00	0.00	60.00	37,227.00
Average Time between j views (1st half)	0.66	1.29	0.20	0.00	12.00	38,455.00
Average Time between j views (2nd half)	1.09	1.74	0.40	0.00	12.00	37,227.00

very small increases afterward. We use these cluster assignment to create three features measuring the percentage of videos that an $artist_i$ has in each of the clusters.

3.4. Early Adoption Features

Features in this group capture temporal patterns of early adoption, by measuring the average time it takes an artist to reach an initial set of views. Research shows that features based on the rate of early adoption have proven to be highly predictive of future popularity. Shulman et al. (2016) found that one can easily predict whether digital items (spanning music, books, photos, and URLs) will end up in the top half percentile of popular items using speed of adoption alone. Similarly, Cheng et al. (2014), found that similar temporal features are important predictors for Facebook posts' popularity. We hypothesize that videos with higher rates of early adoption are associated to artists that are able to quickly engage an audience and can therefore be predictive of future success. For each of the videos in our data set, we compute:

- Time to j views: the days between adoption of initial viewers and adoption of j views (where $100 < j < k$)
- $time_{1..k/2}$: Mean days between adoptions for the first half of the adoptions.
- $time_{k/2..k}$: Mean days between adoptions for the last half of the adoptions.

where k is set to be 1,000 views. Around 72 percent of the videos in our data set reached this threshold in the first 60 days. Summary statistics are shown for this metrics in Table 3. Given that we make predictions at an artist level, we summarized the metrics for each artist by computing their averages:

- $avgTime_j$: Average days to j viewers for $artist_i$
- $avgTime_{1..k/2}$: Average of $time_{1..k/2}$ for $artist_i$
- $avgTime_{k/2..k}$: Average of $time_{k/2..k}$ for $artist_i$

3.5. Additional Features

Along with the metrics listed in the previous sections, we also included features in our model related to the video production capacity measured by computing the artist's average uploads per

month; the percentage of album releases belonging to a given genre for each artist; and their the best and worst performing video in terms of views. In Table 4 we show summary statistics of for all the metrics discussed in the previous sections, and a detailed description of each metric is included in the Appendix in Table A1 .

4. Empirical Analysis

As a first analysis, we'll use a simple logistic regression to show that features derived from YouTube videos describing an artist's performance in the platform are significantly associated with our two metrics of success. We also compute the R^2 coefficient in order to measure the proportion of variation in the dependent variable explained by the predictors. In the next section, we leverage machine learning algorithms that place less assumptions on the underlying data to predict the two measures of success. We identify the features that are most important in making predictions and evaluate our results against an external metric of success.

4.1. Analysis using a simple Logistic Regression Model

We formulate the logit model as follows:

$$P(Y_{success} = 1|\cdot) = F(\beta_0 + \beta_1 AverageMetrics_i + \beta_2 PercentChange_i + \\ \beta_3 ClustersPercentage_i + \beta_4 TimeToJViews_i + \beta_5 AvgNumberUploads_i + e_i)$$

Where $Y_{success}$ is the dependent variable measuring the metrics of success for $artist_i$. As outcome of interest, we'll use $Signed_i$, indicating whether the artists will release an album with a major record label in the following period, and $Chart_i$, indicating whether the artist has a first-time Billboard appearance. The video features are grouped into their overarching category as described in sections 3. $AverageMetrics_i$ corresponds to the features described in the Average Audience Engagement Features section, $PercentChange_i$ to the percentage change features described in the Video Performance Percentage Change section, $ClustersPercentage_i$ corresponds to the features describing the percentage of videos belonging to each view trends cluster as described in the Cluster Features section, and $TimeToJViews_i$ groups the features describing the average time to j views for each $artist_i$, as described in the Early Adoption section. We also include $AvgNumberUploads_i$, indicating the average number of videos uploaded to YouTube per month within our time window of observation.

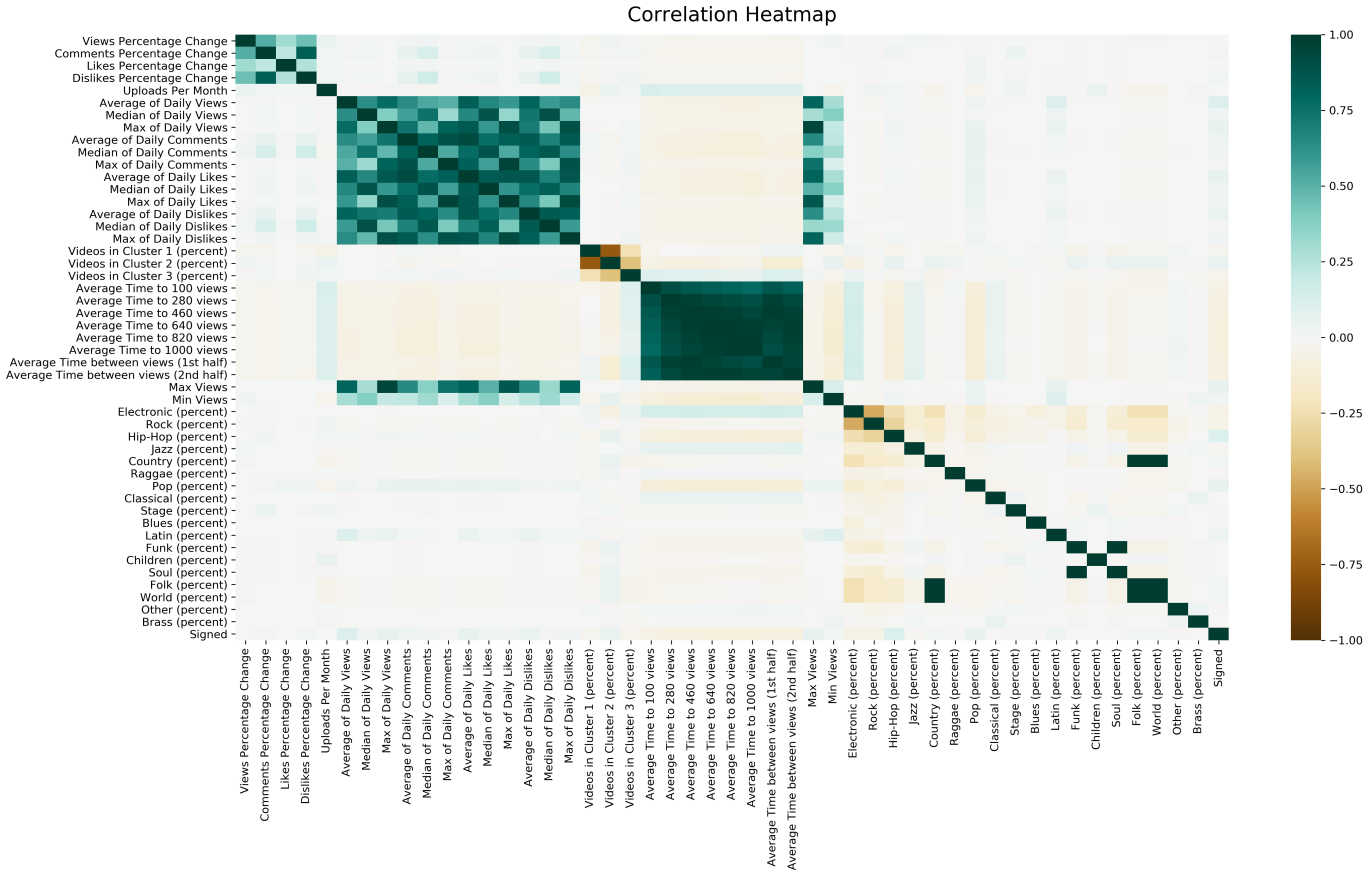
4.2. Releasing an album with a major record label

We first set our metric of success to $Y_{success} = Signed_i$, a variable indicating whether the artist released an album with a major record label in the next period (calendar year) for the first time in their career. The results from the full specification logit model are shown in Table 5, column one.

Table 4 Summary Statistics for features summarizing the artists performance in YouTube. Statistics are taken across artists, as opposed to videos as in Tables 1 - 3

Feature Name	Mean	Std. Dev.	Median	Min	Max
Views Percentage Change	0.852	18.663	0.003	-0.978	1,365.123
Comments Percentage Change	0.126	2.367	0.003	-0.989	117.621
Likes Percentage Change	0.461	14.095	0.003	-0.989	1,034.481
Dislikes Percentage Change	0.293	7.539	0.002	-0.989	488.659
Uploads Per Month	3.059	6.460	1.375	0.167	208.800
Average of Daily Views	543.149	7,987.180	14.756	0.015	451,533.432
Median of Daily Views	252.686	4,154.926	8.408	0.006	252,437.935
Max of Daily Views	3,000.061	49,569.034	41.318	0.022	3,161,012.857
Average of Daily Comments	1.265	13.363	0.144	0.004	612.985
Median of Daily Comments	0.594	5.596	0.072	0.003	295.246
Max of Daily Comments	8.008	150.064	0.500	0.004	9,612.286
Average of Daily Likes	13.123	162.431	0.550	0.004	8,332.370
Median of Daily Likes	6.562	82.507	0.355	0.004	3,829.645
Max of Daily Likes	80.783	1,727.120	1.542	0.004	124,095.571
Average of Daily Dislikes	0.526	7.527	0.066	0.004	428.798
Median of Daily Dislikes	0.226	3.099	0.040	0.003	208.004
Max of Daily Dislikes	3.219	64.714	0.220	0.004	4,158.000
Videos in Cluster 1 (percent)	0.405	0.401	0.333	0.000	1.000
Videos in Cluster 2 (percent)	0.458	0.426	0.400	0.000	1.000
Videos in Cluster 3 (percent)	0.124	0.272	0.000	0.000	1.000
Average Time to 100 views	18.819	25.155	2.000	0.000	61.000
Average Time to 280 views	24.019	26.570	6.000	0.000	61.000
Average Time to 460 views	26.378	26.618	10.000	0.000	61.000
Average Time to 640 views	28.038	26.396	14.000	0.000	61.000
Average Time to 820 views	29.481	26.111	18.000	0.000	61.000
Average Time to 1000 views	30.920	25.736	22.000	0.000	61.000
Average Time between j views (1st half)	23.760	27.951	2.200	0.000	61.000
Average Time between j views (2nd half)	26.248	28.034	5.000	0.000	61.000
Max Views	206,202.101	4,359,970.025	2,378.000	0.000	276,175,983.000
Min Views	14,244.480	111,881.274	1,251.000	0.000	3,141,957.000
Electronic (percent)	0.305	0.413	0.000	0.000	1.000
Rock (percent)	0.298	0.412	0.000	0.000	1.000
Hip-Hop (percent)	0.140	0.320	0.000	0.000	1.000
Jazz (percent)	0.031	0.145	0.000	0.000	1.000
Country (percent)	0.027	0.079	0.000	0.000	0.333
Raggae (percent)	0.008	0.077	0.000	0.000	1.000
Pop (percent)	0.060	0.164	0.000	0.000	1.000
Classical (percent)	0.013	0.095	0.000	0.000	1.000
Stage (percent)	0.008	0.067	0.000	0.000	1.000
Blues (percent)	0.009	0.068	0.000	0.000	1.000
Latin (percent)	0.005	0.053	0.000	0.000	1.000
Funk (percent)	0.017	0.069	0.000	0.000	0.500
Children (percent)	0.001	0.018	0.000	0.000	0.917
Soul (percent)	0.017	0.069	0.000	0.000	0.500
Folk (percent)	0.027	0.079	0.000	0.000	0.333
World (percent)	0.027	0.079	0.000	0.000	0.333
Other (percent)	0.006	0.053	0.000	0.000	1.000
Brass (percent)	0.000	0.005	0.000	0.000	0.221

Figure 3 Correlation Matrix of video features. Dark blue colors indicate higher correlation. We notice



We observe that only one variable has a significant association with whether an artist will work with a major label. However, there is multicollinearity between most variables within the same overarching group, as shown in the correlation matrix in Figure 3. In contrast, Figure 3 also shows that correlation is not as strong across groups of variables. Therefore, we run a specification with only one variable from each group of features, as defined in section 3. The results are shown in Table 5 column two, where we observed that the coefficients for "Videos of Cluster 2 percent", "Average Time to 1000 Viewers" and "Average of Daily Views" are significant and in the direction we would expect. We observe a strong positive association between "Videos in Cluster 2 percent"—which describes the percentage of videos that an artist has that belong to "Cluster 2"—and the likelihood of working with a major label in the next calendar year. This could be driven by videos in "Cluster 2" having a higher and non-slowing rate of cumulative views. Similarly, we observe a significant negative coefficient for "Time to 1000 views", indicating that taking a longer time to reach 1000 views in average is negatively correlated with working with a major record label. We are careful not to draw strong conclusions from the size of the coefficients, given that all our features were derived from the same set of initial metrics and correlation may persist even within the reduced set, and there might be unobservable variables.

Table 5 Results of a Logistic Regression model using $Signed_i$ as dependent variable

	Dependent variable:	
	Release with Major Record Label Next period	
	(1) Full Specification	(2) One Variable per Group
Views Percentage Change	0.0001 (0.010)	−0.001 (0.006)
Comments Percentage Change	−0.0004 (0.001)	
Likes Percentage Change	−0.00001 (0.0001)	
Dislikes Percentage Change	0.0001 (0.0001)	
Uploads Per Month	0.010 (0.008)	0.008 (0.008)
Average of Daily Views	0.0001* (0.0001)	0.00001*** (0.00001)
Median of Daily Views	−0.00001 (0.0001)	
Max of Daily Views	0.00000 (0.00001)	
Average of Daily Comments	0.035 (0.083)	
Median of Daily Comments	−0.002 (0.084)	
Max of Daily Comments	−0.009 (0.009)	
Average of Daily Likes	−0.006 (0.008)	
Median of Daily Likes	0.002 (0.008)	
Max of Daily Likes	0.001* (0.001)	
Average of Daily Dislikes	0.068 (0.169)	
Median of Daily Dislikes	−0.207 (0.302)	
Max of Daily Dislikes	−0.027* (0.016)	
Videos in Cluster 1 (percent)	0.737 (0.617)	
Videos in Cluster 2 (percent)	1.872*** (0.580)	1.094*** (0.246)
Average Time to 100 views	−0.014 (0.017)	
Average Time to 280 views	0.006 (0.041)	
Average Time to 460 views	−0.083 (0.093)	
Average Time to 640 views	−0.013 (0.084)	
Average Time to 820 views	0.031 (0.099)	
Average Time to 1000 views	−0.052 (0.058)	−0.036*** (0.006)
Average Time between views (1st half)	0.053 (0.054)	
Average Time between views (2nd half)	0.034 (0.033)	
Max Views	0.000 (0.00000)	
Min Views	−0.00000 (0.00000)	
Constant	−4.526*** (0.555)	−3.933*** (0.195)
Observations	6,259	6,259
Log Likelihood	−484.030	−501.184
Akaike Inf. Crit.	1,028.060	1,014.367
R^2	0.049	0.023
Note:		*p<0.1; **p<0.05; ***p<0.01

Changing the variable selected from each group does not change the results in terms of significance. In two alternative specifications, we include "Videos in Cluster 3 (percent)" and "Videos in Cluster 1 (percent)" in place of Cluster 2 (see Appendix A4). We observe significant coefficients in both cases, but the direction changes: a higher number of videos in Clusters 1 and 3—which

group videos with concave shapes where rate of viewership decreases quickly after the first 10-28 days—are associated with a lower likelihood of working with a major record label.

To understand how well the logistic regression fits the data, we computed the R^2 coefficient. The R^2 statistic tells us how well the model and our set of predictors explain the variability of the dependent variable, and is a well-defined measure used in linear regressions. Different extensions of the R^2 to generalized linear models have been developed. We use the coefficient of determination R^2 as defined by Zhang (2017), which uses the variance function to define the total variation of the dependent variable as well as the remaining variation of the dependent variable after modeling the predictive effects of the independent variables.

The results from the R^2 analysis show that the features as modeled in the logistic regression explain only 4.9 percent of the variance of the dependent variable, indicating that a simple logit model may not be suitable for our prediction task.

4.3. Appearance in a Billboard chart

We replicate the previous analysis, this time using $Y_{success} = Chart_i$ as our dependent variable, indicating whether $artist_i$ appeared in the Billboard charts in the next period (calendar year) for the first time ever. Results of the full specification are shown in Table 6 column one. We observe a higher number of variables that have a significant association with the target variable, as compared to the previous analysis using $Signed_i$ as dependent variable. Specifically, variables measuring average uploads per month, average number of dislikes, percentage change in views and dislikes have a significant coefficient. However, we still face multicollinearity within variables of the same group, so we select a subset of features from each group and report the coefficients in Table 6 column two. All the coefficients in this second specification have a significant association with the target variable. Again, "Videos in Cluster 2 percent" shows a strong positive correlation with the dependent variable, indicating that artists that post videos with non-slowing rate of cumulative views in the first 2 months after posting are more likely to appear on Billboard. Additional specifications replacing the variable from each group are shown in the Appendix in A5

The resulting R^2 for the logistic regression is 0.0834, indicating that the feature groups explain a total of 8.5 percent of the variation of the dependent variable of whether an artist will appear in Billboard in the next period.

Comparing the results from the logistic regression and the R^2 coefficients between the models with the two different metrics of success suggests that our features are better predictors of appearance in Billboard charts than they are of an artist working by a major record label. However, both models fail to explain most of the variation in the dependent variables. The relations between the features and the dependent variable may be better modeled assuming less constrains in the data by using Machine Learning methods, as we demonstrate in the next section.

Table 6 Results of a Logistic Regression model using $Chart_i$ as dependent variable

	Dependent variable:	
	Billboard Appearance Next Period	
	(1) Full Specification	(2) One Variable per Group
Views Percentage Change	0.010** (0.004)	0.007** (0.003)
Comments Percentage Change	0.00002 (0.0004)	
Likes Percentage Change	0.00002 (0.00002)	
Dislikes Percentage Change	0.001*** (0.0004)	
Uploads Per Month	−0.126*** (0.049)	−0.104** (0.043)
Average of Daily Views	0.0004** (0.0001)	0.0001*** (0.00002)
Median of Daily Views	−0.0003* (0.0001)	
Max of Daily Views	−0.00003 (0.00002)	
Average of Daily Comments	−0.030 (0.088)	
Median of Daily Comments	0.004 (0.104)	
Max of Daily Comments	−0.001 (0.009)	
Average of Daily Likes	0.011 (0.009)	
Median of Daily Likes	0.006 (0.009)	
Max of Daily Likes	−0.001 (0.001)	
Average of Daily Dislikes	−0.744*** (0.268)	
Median of Daily Dislikes	0.044 (0.359)	
Max of Daily Dislikes	0.079*** (0.029)	
Videos in Cluster 1 (percent)	−0.095 (0.369)	
Videos in Cluster 2 (percent)	0.646* (0.354)	0.579*** (0.198)
Average Time to 100 views	−0.008 (0.012)	
Average Time to 280 views	−0.007 (0.024)	
Average Time to 460 views	0.007 (0.071)	
Average Time to 640 views	0.012 (0.076)	
Average Time to 820 views	−0.080 (0.065)	
Average Time to 1000 views	0.013 (0.030)	−0.025*** (0.004)
Average Time between views (1st half)	0.001 (0.038)	
Average Time between views (2nd half)	0.035* (0.021)	
Max Views	0.00000 (0.00000)	
Min Views	−0.00000 (0.00000)	
Constant	−3.034*** (0.331)	−3.055*** (0.163)
Observations	5,297	5,297
Log Likelihood	−606.130	−632.384
Akaike Inf. Crit.	1,272.260	1,276.768
R^2	0.084	0.043
Note:		*p<0.1; **p<0.05; ***p<0.01

5. Machine Learning classifiers to predict artist success

A simple logistic regression model analysis showed that there are statistically significant associations between the YouTube video features and the dependent variables ($Signed_i$ and $Chart_i$). In this section, we use machine learning methods that place less assumptions on the underlying data to predict both measures of success and evaluate their performance in an unseen test data set. By

performing a permutation feature importance analysis, we show that average performance metrics and growth in the platform are the most important features used by the models to predict both metrics of success. Finally, we compare our ranked predictions to an external data set collected from Spotify, that measures artist popularity and number of followers.

5.1. Task: Predict releasing an album with a major record label

To simulate the conditions under which an A&R department would take the decision to sign an artist, we set an arbitrary cutoff date of access to data to train our model, to 2019-01-01. Our training data set consists of artists that have YouTube videos in 2017, with the target variable taking value of 1 if the artist was signed by a major record label in 2018 and 0 otherwise. Similarly, our test set consists of artists with YouTube videos in 2018, and the target variable takes the value of 1 if an artist was signed in 2019, and 0 otherwise. There are artists that have videos in both 2017 and 2018, so we randomly assigned them to the test or train data set to avoid evaluating the models with the same set of artists.

With the objective of making predictions on a set of artists from which we have enough information, we only consider artists with at least 2 videos posted during the observation period, where each video metric is observed at least 5 or more times. This reduces the initial set of artists from 35,791 artists to the final data set of 6,259 artists. In total, our final data set is composed of 6,151 artists labeled as 0 (unsigned), and 108 labeled as 1 (signed). This means that only 1.7 percent of the artists are signed by a major record label, which aligns with the expectation that only a small number of artists succeed but showcases the difficulty of the task due to high class imbalance.

We train standard machine learning classifiers for our task: Random Forest (Breiman 2001), XGBoost (Friedman 2001, Chen and Guestrin 2016), Support Vector Machines (Cortes and Vapnik 1995), Gaussian Naive Bayes and Logistic Regression. We perform hyper-parameter tuning via grid search using 5-fold cross validation on the training set, using Average Precision to optimize the classifiers.

We are interested in a model that achieves a high accuracy within the set of artists that are predicted to be signed, which is measured by the model's Precision. Such model would give confident predictions, allowing a record label to allocate resources in artists with the highest likelihood of becoming successful. Similarly, a second desired business outcome is to capture as many to-be signed artists as possible, which is measured by the true positive rate (or recall). There is almost always a trade off between the two measures, given that as we try to capture more to-be signed artists (i.e increase the recall), the classifier may suffer from a loss of precision. Therefore, we report the Precision-Recall for each of our models, and optimize towards Average Precision. For Average Precision, we use the implementation from Pedregosa et al. (2011), that summarizes a

precision-recall curve as the weighted mean of precisions achieved at each threshold. This is our main outcome of interest to measure the performance of the classifiers.

To address class imbalance, we compute the average Precision-Recall curve, obtained by performing 5-fold cross validation in our training data set, and select a decision threshold that achieves a Precision of at least 10 percent. We replicate this process for each classifier and report the performance using the cross-validated probability threshold in the unseen data set. As an alternative for handling class imbalance, we tried SMOTE (Synthetic Minority Over-sampling Technique) as described by Chawla et al. (2002), an over-sampling approach in which the minority class is over-sampled by creating "synthetic" examples rather than by naive oversampling with replacement. This generates a more balanced data set. However, the models over-fit to the training data and perform poorly in the test data set (see Appendix A6).

We also compute the ROC curve and the AUC (area under the ROC curve) for each classifier, given that we are interested in a model that achieves a high true positive rate (TPR), with a low false positive rate (FPR). A high TPR and a low FPR indicate that our model can identify a significant percentage of artists who will get signed, without capturing a large percentage of the unsigned artists in the process.

Another way to measure the performance of the classifiers is by providing a ranking based on the models' predicted probability of success, and then comparing the ranking against the ground truth labels. Ideally, artists high in the ranking would be more likely to have a positive label (i.e. they are end up being signed artists). To show this, we produce a ranking for each classifier and plot the ranking position against the percentage of artists up to position n that were correctly identified as eventually signed, for n in $\{1...\text{TestSetSize}\}$. We are particularly interested on predictions in the top of the ranking, so we compute the curve for the predicted top 200 artists, and report them in the plot labeled "Top 200 Ranking".

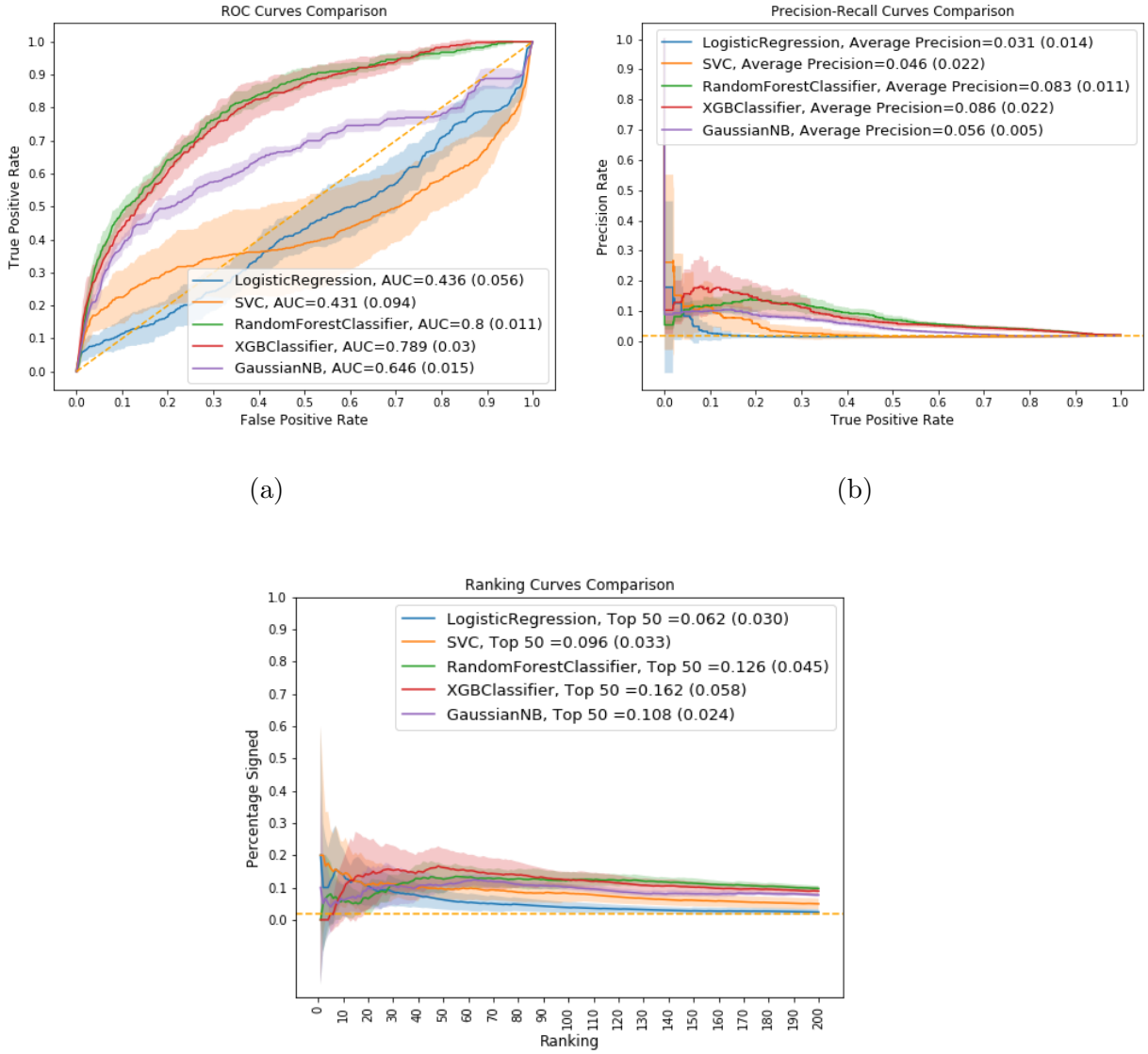
Given that there is random assignation for some of the artists to the train and test data sets, we run each model 10 times. In average, the test set was composed of 2,828 artists, with 50 artists labeled positive and 2,774 negative. In Table 7, we report the mean and standard deviation for each metric. Similarly, we average the test Precision-Recall curve, ROC curve and "Top 200 Ranking" curve and report them in Figures 4a, 4b, and 4c respectively.

As shown in Table 7, the two best performing models are Random Forest and XGBoost, both decision-tree-based methods. In particular, XGBoost is able to achieve an Average Precision of 8.6 percent, Precision of 15 percent, and ROC AUC of 79 percent. Random Forest achieves similar metrics, in addition to achieving a higher True Positive Rate of 38 percent and generating more stable results. In average, among the artists predicted to be signed by the two best models, 10-15 percent are correctly classified, and we are able to capture 25-38 percent of the total to-be signed

Table 7 **10-Run average results for predicting $Singed_i$, whether an artist will work with a major record label in the next period. We report the average and standard deviation for each metric.**

classifiers	AUC	Avg. Precision	Precision	Recall	FPR	F1	Accuracy
LogisticRegression	0.436 (0.056)	0.031 (0.014)	0.017 (0.008)	0.142 (0.074)	0.158 (0.071)	0.03 (0.014)	0.83 (0.069)
SVC	0.431 (0.094)	0.046 (0.022)	0.086 (0.091)	0.251 (0.281)	0.279 (0.412)	0.068 (0.048)	0.713 (0.399)
RandomForestClassifier	0.8 (0.011)	0.083 (0.011)	0.099 (0.023)	0.383 (0.112)	0.071 (0.038)	0.151 (0.021)	0.919 (0.035)
XGBClassifier	0.789 (0.03)	0.086 (0.022)	0.149 (0.088)	0.285 (0.093)	0.042 (0.026)	0.166 (0.037)	0.946 (0.024)
GaussianNB	0.646 (0.015)	0.056 (0.005)	0.098 (0.024)	0.189 (0.052)	0.034 (0.014)	0.121 (0.013)	0.952 (0.012)

Figure 4 **10-Run Average of Precision-Recall curve (a), the ROC curve (b) and the "Top 200 Ranking" curve (c) for predicting $Signed_i$. Shaded regions shows 1 standard deviation from the mean curve. The dotted yellow line in each plot shows the result for a random guess.**



artists. The average Precision-Recall curves in Figure 4b highlight the difficulty of the classification task. The classifiers are not able to achieve a precision greater than 20 percent while also preserving a FPR greater than 10 percent. Finally, inspecting the "Top 200 Ranking" average curves in Figure 4c reveals that around 12-16 percent of the top 50 artists from the two best models' rankings ended up actually being signed.

In summary, predicting whether an artist will work with a major record label using features summarizing the artist's performance in YouTube proves to be a difficult task. In the next subsection, we use the same group of features to predict an arguably more objective measure of success, whether an artist will appear on the Billboard charts in the next period.

5.2. Task: Predict appearance in a Billboard chart

In this section, we discuss the results of using the same group of features to predict a second metric of success: whether the artists will appear in the Billboard charts anytime in the next period (calendar year) for the first time ever. We show that features derived from YouTube are more suitable for making predictions on whether an artist breaks into popularity charts than predicting whether they get to work with a major label.

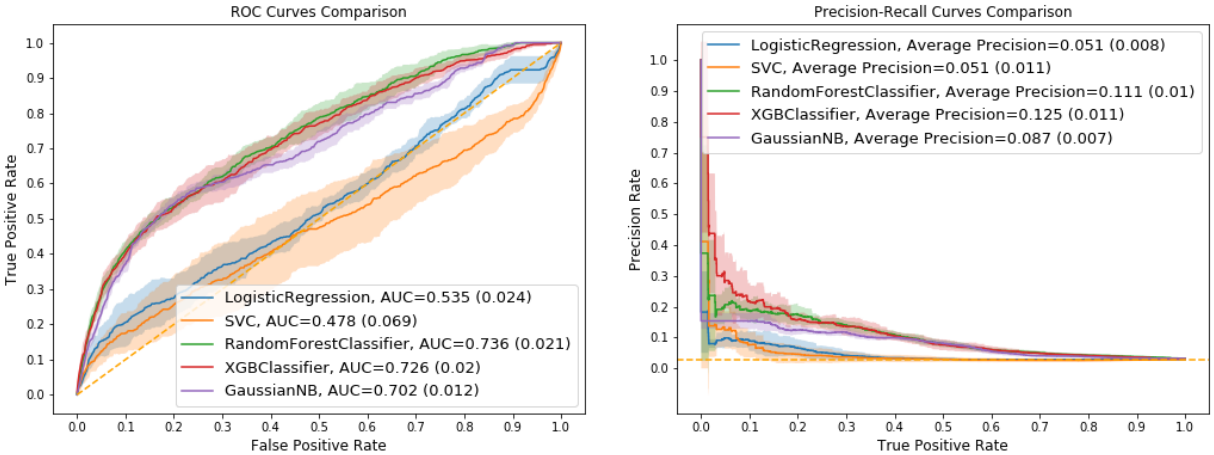
The train set consists of artists with YouTube videos in 2017 who have never appeared on Billboard before 2018-01-01. We test the classifiers' performance on a set of artists that have videos on 2018 and have never appeared on Billboard before 2019-01-01. Since there are artists that have videos in both years 2017 and 2018, we randomly assigned them to the test or train data set to avoid an overlap of artists between the data sets. Almost 3 percent of artists in our data set appeared on Billboard in the following year, making this a classification problem with high class imbalance, albeit slightly less imbalanced than predicting $Signed_i$.

We use the same set of classifiers as described in the previous section and follow the same training procedure to select hyper-parameters and to deal with class imbalance. We report the models' average performance for 10 runs in Table 8, along with the resulting average ROC curves, Precision-Recall curves and "Top 200 Ranking" curves in Figures 5a, 5b, 5c respectively.

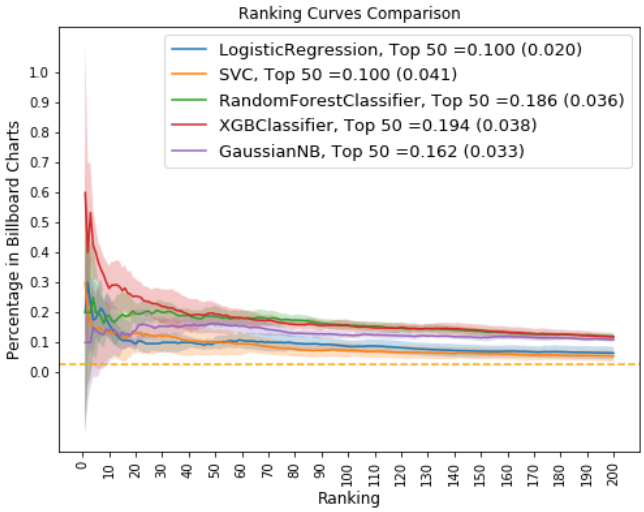
Table 8 10-Run average results for predicting $Chart_i$, whether an artist will appear in a Billboard chart in the next period. We report the average and standard deviation for each metric.

classifiers	AUC	Avg. Precision*	Precision	Recall	FPR	F1	Accuracy
LogisticRegression	0.535 (0.024)	0.051 (0.008)	0.109 (0.048)	0.103 (0.054)	0.025 (0.013)	0.097 (0.046)	0.95 (0.011)
SVC	0.478 (0.069)	0.051 (0.011)	0.096 (0.065)	0.361 (0.349)	0.323 (0.413)	0.088 (0.035)	0.668 (0.391)
RandomForestClassifier	0.736 (0.021)	0.111 (0.01)	0.189 (0.023)	0.079 (0.023)	0.01 (0.004)	0.108 (0.023)	0.963 (0.003)
XGBClassifier	0.726 (0.02)	0.125 (0.011)	0.334 (0.121)	0.055 (0.024)	0.004 (0.003)	0.09 (0.033)	0.969 (0.003)
GaussianNB	0.702 (0.012)	0.087 (0.007)	0.151 (0.022)	0.146 (0.041)	0.026 (0.01)	0.143 (0.015)	0.95 (0.009)

Figure 5 **10-Run Average of Precision-Recall curve (a), the ROC curve (b) and the "Top 200 Ranking" curve (c) for predicting $Chart_i$. Shaded regions shows 1 standard deviation from the mean curve. The dotted yellow line in each plot shows the result for a random guess.**



(a) (b)



(c)

Once again, Random Forest and XGBoost achieve the best results. The two best performing models are able to achieve an Average Precision between 11 and 12.5 percent, a Precision of 19-34 percent and an AUC of 72-74 percent. In contrast to predicting $Signed_i$, we notice that for this classification task, the models are able to achieve better performance in terms of precision and recall as seen in the "Precision-Recall" curves in Figure 5b. Particularly, XGBoost and Random Forest are able to achieve precision values slightly below 20 percent while maintaining a True Positive rate close to 20 percent. This means that among the artists predicted to be appear on

Billboard, close to 20 percent are correctly classified, and we are able to capture around 20 percent of the total artists that ended up breaking into the charts. Similarly, the "Top 200 Ranking" curve in Figure 5c shows that around 19 percent of the artists in the top 50 rank generated from the classifiers made it to the Billboard charts in the next period.

The results suggest that features based on YouTube video metrics are more suitable for making predictions on whether an artist break into popularity charts for the first time than predicting whether an artists gets to work with a major label. To further explore this result, we set up a task to predict appearance in Billboard charts in the next period, regardless if it's the first appearance. The 10-Run average results for this approach are shown in the Appendix. Notably, the top 50 Rank is able to capture close to 40 percent of artists that make it into the charts the next year.

5.3. Feature Importance

Now, we are interested in understanding which group of features are most important when predicting both measures of success. We show that the best performing model relies primarily on "Average of Video Metrics" features for both classification tasks. To assess feature importance, we'll use the permutation feature importance algorithm (Breiman 2001, Molnar 2019). This algorithm measures the increase in the prediction error of the model (or any other metric to assess model performance) by breaking the relationship between the features of interest and the target variable. This is achieved by permuting the feature's values and measuring the drop in performance.

Given that we trained our models to optimize Average Precision, we will measure the decrease in Average Precision when we permute each feature in our model. The presence of correlated features complicates the interpretation of the method. Correlated features may decrease the importance of the associated feature since the importance will be spitted between the correlated features. This could lead to an incorrect interpretation of a ranking of features based on the importance score. Our data set contains sets of highly correlated features, as shown in the correlation matrix in Figure 3. To address this, we implement a modified version of the the permutation algorithm that allows permutation of correlated group variables (Gregorutti et al. 2015), using the features groups we defined in section 3. We present the average permutation results for the XGBoost model—the best performing classifier in terms of Average Precision— in Tables 9 and Table 10 for predicting $Signed_i$ and $Chart_i$ respectively. We included the grouped feature importance for the remaining models in the Appendix.

The result of the permutation analysis shows that for both classification tasks, Average Precision drops the most when we permute the "Average of Video Metrics" Features. The second most important group of features for predicting $Signed_i$ is "Video Cluster Percentage", although the contribution is small. For the task of predicting Billboard appearance, we observe that the "Video

Table 9 Results from the Permutation Feature Importance Analysis for predicting $Signed_i$. We report the mean and standard deviation for the feature importance score from permuting the group of features 30 times. We include the Average Precision baseline score for the test set.

Feature Group	Mean Feature Importance*	Std. Dev.	Baseline Score
Average of Video Metrics	0.064	0.013	0.116
Video Cluster Percentage	0.031	0.014	0.116
Genres	0.029	0.017	0.116
Video Metrics Percentage Change	0.018	0.011	0.116
Min-Max Views	0.009	0.022	0.116
Uploads per Month	0.000	0.000	0.116
Average Time to J Views	-0.003	0.007	0.116

* Mean Feature Importance measures the average drop in average precision after permuting each group of variables.

Table 10 Results from the Permutation Feature Importance Analysis for predicting $Chart_i$. We report the mean and standard deviation for the feature importance score from permuting the group of features 30 times. We include the Average Precision baseline score for the test set.

Feature Group	Mean Feature Importance*	Std. Dev.	Baseline Score
Average of Video Metrics	0.078	0.007	0.119
Video Metrics Percentage Change	0.022	0.012	0.119
Min-Max Views	0.003	0.006	0.119
Genres	0.001	0.000	0.119
Average Time to J Views	-0.000	0.000	0.119
Uploads per Month	-0.002	0.005	0.119
Video Cluster Percentage	-0.004	0.008	0.119

* Mean Feature Importance measures the average drop in average precision after permuting each group of variables.

Metrics Percentage Change" group—related to the artist's growth in terms of views,likes,dislikes and comments—is the second most important group of features. In both cases, the rest of the feature groups increase Average Precision marginally or are not important for the prediction tasks.

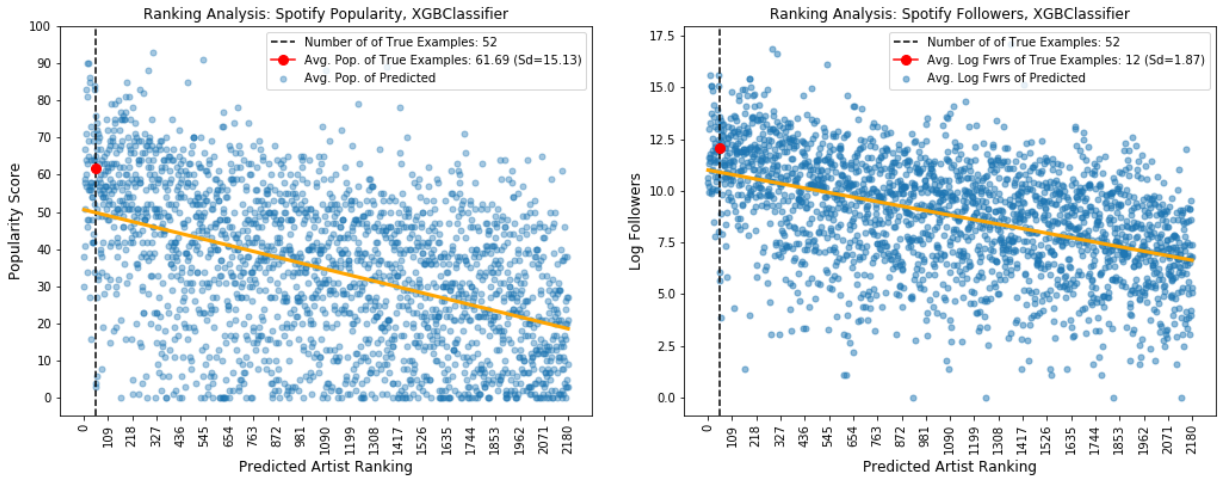
5.4. Ranking Evaluation using a external source: Spotify Popularity Data

In this section, we discuss how the models' rankings perform using an alternative measure of success, based on an artist's popularity on Spotify. Given that our models are built on features that signal popularity on YouTube, we expect that our predicted rankings will be positively correlated with signals of popularity in other steaming platforms.

We collected the Spotify popularity index and number of Spotify followers for each artist in our data set, using the Spotify Web API [17]. The Spotify popularity index is a score computed for all artists in Spotify, that summarizes their popularity on the platform. The score takes a value between 0 and 100, with 100 being the most popular. The artist's popularity is calculated using the popularity of all the artist's tracks, and weights content based on recency. The number of

[17] https://developer.spotify.com/documentation/web-api/

Figure 6 Comparison of ranking produced by XGBoost Classifier in the $Signed_i$ prediction task against Spotify Popularity (a) and Spotify Log Followers (b)



(a) (b)

followers measures how many Spotify users opt in following the artist's profile on Spotify. Both metrics are an alternative and arguably objective measure of popularity, in one of the most widely used streaming platforms in the United States[18].

We ranked the artists in our test set based on their probability of success as given by the models trained in the previous section, and plot the ranked artist's position against their Spotify popularity index and their number of log followers. For reference, we compute the mean Spotify Popularity index and the mean log followers for the ground truth examples. The plots for the $Signed_i$ and the $Chart_i$ classification rankings are shown in Figures 6 and 7 respectively.

By inspecting the plot in Figure 6a, we notice that there is a small but positive relation between a high ranking position and a high popularity index. The plot in Figure 6b also suggests a correlation between a high ranking position and a high number of followers. We expect number of followers to be a more stable metric than popularity index, given that the latter depends highly on recent streaming activity. To verify if the relation between the rank and both Spotify metrics are significant, we regressed the popularity index and the log followers on the ranking position for XGBoost and Random Forest, the best performing models. The regression results for the $Signed_i$ classification ranking are shown in Tables 11 and 12.

The results from the regressions show a significant positive association between a high rank and a high popularity index, as well as a high number of log followers. For the case of the rankings produce from the $Signed_i$ classification task, the XGBoost ranking shows the highest correlation

[18] https://yougov.co.uk/topics/media/articles-reports/2021/03/18/services-used-stream-music-poll

Table 11 **Regression of Spotify Popularity index on Ranking from $Signed_i$ classification Task**

	Dependent variable: Spotify Popularity Index as of March 2021	
	RandomForestClassifier	XGBClassifier
Intercept	48.826***	50.626***
	(0.803)	(0.782)
Predicted Ranking	-0.013***	-0.015***
	(0.001)	(0.001)
Observations	2,180	2,180
R^2	0.161	0.204
Adjusted R^2	0.160	0.203
Residual Std. Error	18.737	18.247
F Statistic	416.559***	557.674***
Note:		*p<0.1; **p<0.05; ***p<0.01

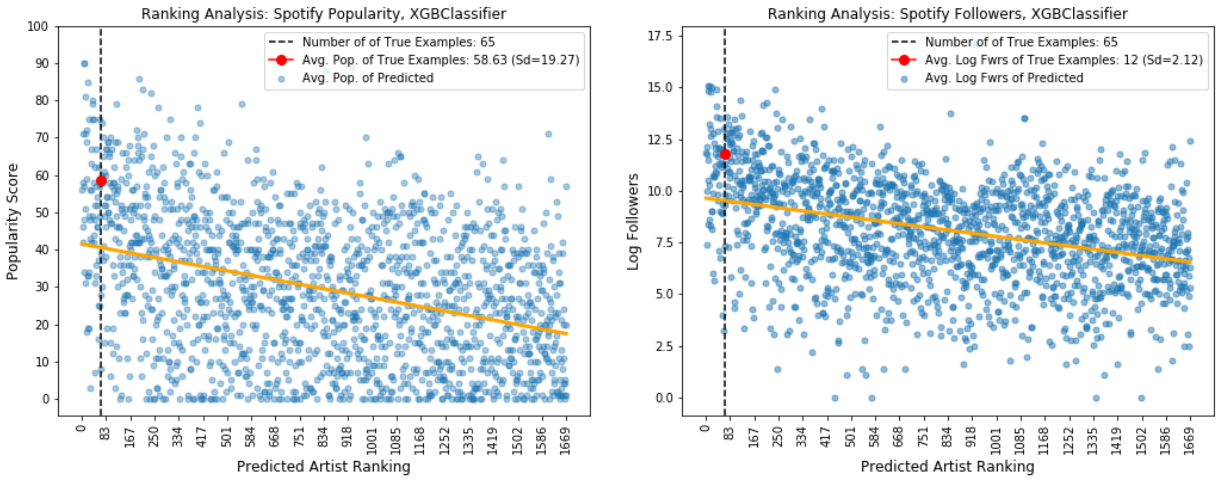
Table 12 **Regression of Spotify Log Followers on Ranking from $Signed_i$ classification Task**

	Dependent variable: Spotify Log Followers as of March 2021	
	RandomForestClassifier	XGBClassifier
Intercept	10.745***	11.001***
	(0.106)	(0.103)
Predicted Ranking	-0.002***	-0.002***
	(0.000)	(0.000)
Observations	2,180	2,180
R^2	0.169	0.217
Adjusted R^2	0.168	0.216
Residual Std. Error	2.466	2.394
F Statistic	441.760***	601.902***
Note:		*p<0.1; **p<0.05; ***p<0.01

Table 13 **Regression of Spotify Popularity index on Ranking from $Chart_i$ classification Task**

	Dependent variable: Spotify Popularity Index as of March 2021	
	RandomForestClassifier	XGBClassifier
Intercept	40.872***	41.497***
	(0.887)	(0.881)
Predicted Ranking	-0.014***	-0.014***
	(0.001)	(0.001)
Observations	1,669	1,669
R^2	0.116	0.130
Adjusted R^2	0.116	0.129
Residual Std. Error	18.114	17.979
F Statistic	219.567***	248.000***
Note:		*p<0.1; **p<0.05; ***p<0.01

Figure 7 **Comparison of ranking produced by XGBoost Classifier in the $Chart_i$ prediction task against Spotify Popularity (a) and Spotify Log Followers (b)**



(a) (b)

Table 14 **Regression of Spotify Log Followers on Ranking from $Chart_i$ classification Task**

	Dependent variable: Spotify Log Followers as of March 2021	
	RandomForestClassifier	XGBClassifier
Intercept	9.703***	9.645***
	(0.114)	(0.114)
Predicted Ranking	-0.002***	-0.002***
	(0.000)	(0.000)
Observations	1,669	1,669
R^2	0.137	0.127
Adjusted R^2	0.136	0.127
Residual Std. Error	2.325	2.338
F Statistic	264.655***	243.387***
Note:		*p<0.1; **p<0.05; ***p<0.01

with the Spotify metrics, where a drop of 20 positions in the ranking is associated with a 0.3 point drop in the popularity index, and a 0.002 drop in log number of followers. For the case of the rankings produce from the $Chart_i$ classification task, we observe that the models' rankings show a significant, but small association with the Spotify metrics as well (see Tables 13 and 14). In the case of the $Chart_i$ rankings, both models' rankings show a similar significant and positive correlation between a high rank and the Spotify metrics, where a drop of 20 positions in the ranking is associated with a 0.28 point drop in the popularity index, and a 0.002 drop in log number of followers.

In summary, we found that there is a significant association between the rankings produced by

the classifiers and the Spotify popularity metrics. However, the correlation is small, an indication of a noisy ranking and/or noisy Spotify metrics. The models were not trained to produce a ranking, rather, they mostly care about classifying the artists into the correct category. An algorithm trained to explicitly produce rankings may show stronger correlations between ranking positions and the Spotify metrics. There is also a significant time factor that might explain the weak correlations. The rankings were created using data from 2018, and the Spotify metrics correspond to March 2021, indicating a 3 year gap between measurements.

6. Discussion

In this research we study how social metrics can be used to predict milestones in an artist career. Leveraging a unique data set of YouTube engagement metrics, musical releases and popularity charts, we attempt to develop a predictive model that A&R departments can use to assist in selecting artists with higher likelihood of success among the enormous number of new musicians, and new content, available online. We look at two metrics of success: the first time an artist releases an album with a major record label, and their first appearance in a Billboard chart. We design features from the artists' YouTube videos, mainly around their average performance in the platform as measured by number of views, likes, dislikes and comments and their viewership patterns. We then apply machine learning algorithms in two classification tasks, aiming to predict our two metrics of success. We show that features derived from YouTube videos are predictive of our success metrics, but are not sufficient to accurately capture a majority of potentially successful artists. The best performing models are able to achieve an average precision of 9 percent for the first task and 12 percent for the second, while a top 50 ranking created with the classifiers' probabilities correctly contains 16 percent of to-be signed artists in the first task, and 19 percent of to-appear on Billboard artists in the second task. Our results suggest that YouTube video features are more predictive of an artist's future appearance on a chart, an arguably more objective measure of success which is driven by audience consumption of the artist's music. A comparison with Spotify metrics describing an artist popularity shows that a high position in the rankings generated from the classifiers is positively correlated with more followers and a higher popularity score, although the association is small.

One limitation of our approach is that we only have data from one social media platform. Even though we expect that performance across platforms is highly correlated, we believe that including multiple sources can improve the classifier performance. Another limitation of our data is that we have a relatively short observation period for each artist. Observing the same artist for multiple years may provide a better picture of their performance and growth in the platform. In addition, we might consider other kinds of information to complement audience metrics. Even though previous

research has shown that audience metrics alone can do a great job predicting popularity (Shulman et al. 2016), content derived features (from videos or songs) could provide signals of quality that can predict future success. Similarly, features derived from text such as comments and tweets, as well as number followers, could provide additional signals of audience engagement.

There are also opportunities to explore how social media impacts an artist career using other measures of success. The decision of predicting whether an artists will be offered the opportunity to work with a record label may be driven by reasons beyond an artist performance in a social platform. There are alternative and more objective career outcomes we can consider that are potentially correlated with an artist's performance online. For example, we could consider the number of concerts and the revenue collected from them. This would be of particular importance to a record label that is interested in knowing if an artist will become a monetary success. Similarly, we could use other proxies for popularity. The Spotify popularity metrics can be leveraged to train a model that predicts how popular an artist will be in Spotify and other streaming platforms. This could serve as an important indicator for labels, given's Spotify position as a major player in musical distribution. Overall, our results motivate the further study of the association between observable social media performance and more objective and readily available measures of success.

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7. Appendix

Table A1 Feature Description

Feature Group	Description
Average Audience Engagement Features	
Average of daily views	Computed by taking the daily average number of views for each video for the period between creation date and the latest available update date, and then taking an average for all videos of $artist_i$.
Average of daily comments	Computed following the same procedure as for views, but for number of comments.
Average of daily likes	Computed following the same procedure as for views, but for number of likes.
Average of daily dislikes	Computed following the same procedure as for views, but for number of dislikes.
Video Performance Percentage Change Features	
Percentage Change in Views	Computed by taking the percentage change between the first observed video's average daily views and the last observed video's average daily views for $artist_i$. We normalize this metric by dividing by the number of days between the posting of the first and last video observed.
Percentage Change in Comments	Computed following the same procedure as for views, but for average daily comments.
Percentage Change in Likes	Computed following the same procedure as for views, but for average daily like.
Percentage Change in Dislikes	Computed following the same procedure as for views, but for average daily Dislikes.
Video Cluster Features	
Videos in Cluster 1 (percent)	Percentage of videos belonging to Cluster 1 for $artist_i$
Videos in Cluster 2 (percent)	Percentage of videos belonging to Cluster 2 for $artist_i$
Videos in Cluster 3 (percent)	Percentage of videos belonging to Cluster 2 for $artist_i$
Early Adoption Features	
Average Time to j Views	Average time between of adoption of j views (where $100 < j < 1000$) for $artist_i$. The average is taken across all the artist's videos.
Average Time between j views (1st half)	Computed by first taking the average time for an $artist_i$'s video to go from i to j views (where $100 < i < j < 500$), and then taking an average for all videos of $artist_i$.
Average Time between j views (2nd half)	Computed by first taking the average time for an $artist_i$'s video to go from i to j views (where $500 < i < j < 1000$), and then taking an average for all videos of $artist_i$.
Additional Features	
Average Upload Per Month	Measures the average number of videos uploaded to YouTube per month by $artist_i$
Max Views	Maximum number of views achieved by an artist in a single video by day 60 after creation date
Min Views	Minumum number of views achieved by an artist in a single video by day 60 after creation date
$Genre_i$ (percent)	Percentage of releases that are tagged as $Genre_i$ for $artist_i$, for the following set of 18 Genres: Electronic, Rock, Hip-Hop, Jazz, Country, Raggae, Pop, Classical, Stage, Blues, Latin, Funk, Children, Soul, Folk, World, Other, Brass

Figure A1 **Alternative Interpolation: Resulting Video Clusters from applying the DTW algorithm to time series of the cumulative views observed for the first 60 days since video creation. For video time series in these clusters, we used monotonic cubic interpolation.**

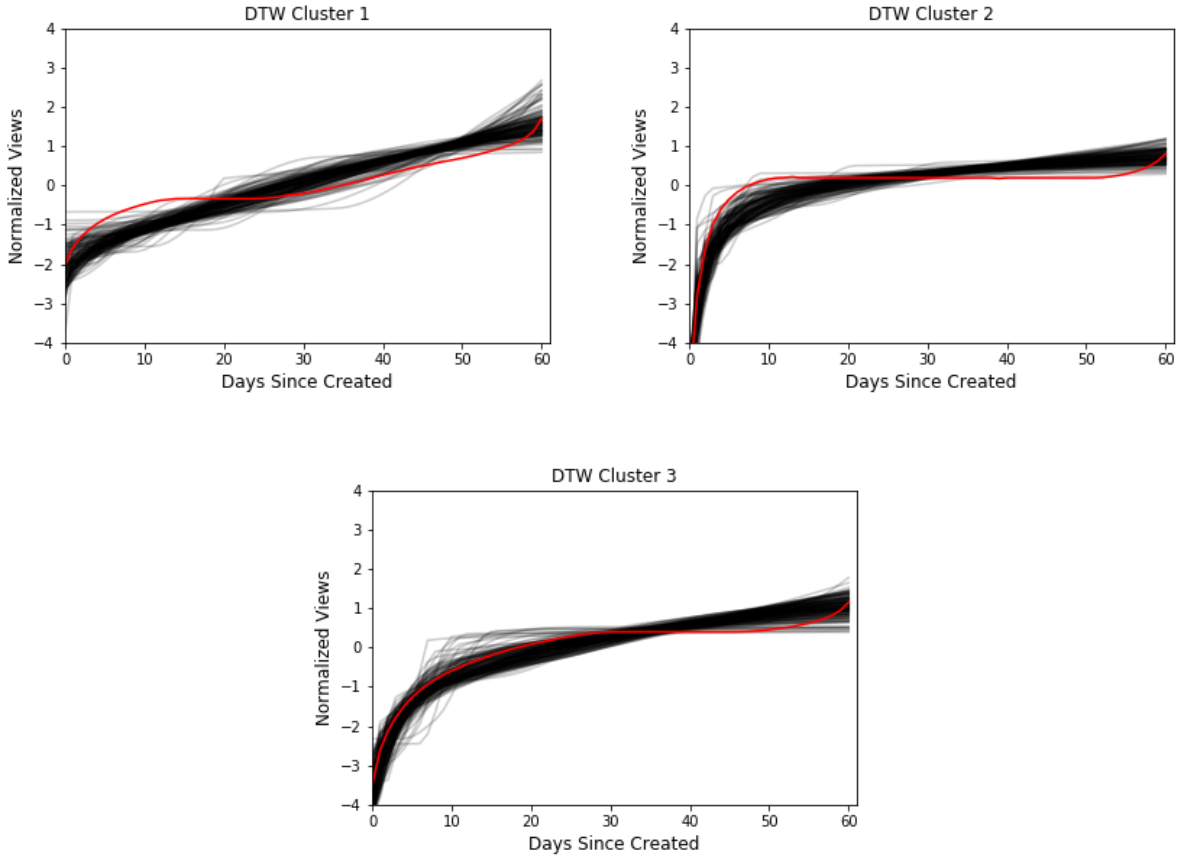


Table A2 **Alternative Interpolation: 10-Run average results for predicting $Signed_i$, whether an artist will work with a major record label in the next period. We report the average and standard deviation for each metric. Time Series used to create Cluster and Time to j features were interpolated using cubic monotonic interpolation.**

classifiers	AUC	Precision	Recall	FPR	Avg. Precision	F1	Accuracy
LogisticRegression	0.457 (0.045)	0.032 (0.046)	0.16 (0.064)	0.165 (0.077)	0.04 (0.013)	0.042 (0.039)	0.823 (0.075)
SVC	0.492 (0.103)	0.067 (0.047)	0.19 (0.156)	0.155 (0.245)	0.044 (0.015)	0.07 (0.045)	0.833 (0.239)
RandomForestClassifier	0.802 (0.014)	0.113 (0.021)	0.3 (0.131)	0.044 (0.021)	0.089 (0.016)	0.158 (0.037)	0.945 (0.018)
XGBClassifier	0.808 (0.025)	0.133 (0.049)	0.329 (0.078)	0.048 (0.022)	0.11 (0.018)	0.174 (0.026)	0.941 (0.021)
GaussianNB	0.644 (0.054)	0.093 (0.029)	0.255 (0.102)	0.073 (0.1)	0.058 (0.01)	0.126 (0.033)	0.915 (0.096)

Table A3 **10-Run average results for predicting $Chart_i$, whether an artist will work with a major record label in the next period. We report the average and standard deviation for each metric. Time Series used to create Cluster and Time to j features were interpolated using cubic monotonic interpolation.**

classifiers	AUC	Precision	Recall	FPR	Avg. Precision	F1	Accuracy
LogisticRegression	0.557 (0.05)	0.105 (0.022)	0.163 (0.104)	0.043 (0.026)	0.059 (0.008)	0.113 (0.042)	0.934 (0.023)
SVC	0.465 (0.098)	0.135 (0.077)	0.095 (0.089)	0.044 (0.082)	0.054 (0.015)	0.083 (0.046)	0.931 (0.078)
RandomForestClassifier	0.742 (0.019)	0.204 (0.039)	0.084 (0.029)	0.01 (0.004)	0.113 (0.012)	0.116 (0.031)	0.963 (0.003)
XGBClassifier	0.74 (0.015)	0.308 (0.134)	0.073 (0.052)	0.007 (0.009)	0.133 (0.016)	0.102 (0.044)	0.966 (0.007)
GaussianNB	0.713 (0.012)	0.167 (0.022)	0.162 (0.038)	0.025 (0.009)	0.096 (0.008)	0.161 (0.019)	0.951 (0.007)

Table A4 Results from Logistic Regression model using $Signed_i$ as dependent variable with alternative specifications

	Dependent variable:			
	Target			
	(1)	(2)	(3)	(4)
Views Percentage Change	−0.0003 (0.005)	−0.001 (0.005)	−0.001 (0.005)	−0.001 (0.006)
Uploads Per Month	0.009 (0.008)	0.007 (0.008)	0.008 (0.009)	0.008 (0.008)
Average of Daily Views	0.00001*** (0.00001)	0.00001*** (0.00000)	0.00002*** (0.00001)	0.00001*** (0.00001)
Videos in Cluster 1 (percent)		−0.795*** (0.272)		
Videos in Cluster 2 (percent)			0.965*** (0.240)	1.037*** (0.244)
Videos in Cluster 3 (percent)	−1.451** (0.584)			
Time to 100 views			−0.035*** (0.008)	
Time to 640 views				−0.036*** (0.006)
Time to 1000 views	−0.034*** (0.005)	−0.035*** (0.005)		
Constant	−3.247*** (0.130)	−3.057*** (0.156)	−4.233*** (0.188)	−4.011*** (0.192)
Observations	6,259	6,259	6,259	6,259
Log Likelihood	−507.311	−507.034	−510.765	−502.914
Akaike Inf. Crit.	1,026.621	1,026.068	1,033.529	1,017.828

Note: *p<0.1; **p<0.05; ***p<0.01

Table A5 Results from Logistic Regression model using $Chart_i$ as dependent variable with alternative specifications

	Dependent variable:			
	Billboard Appearance Next Period			
	(1)	(2)	(3)	(4)
Views Percentage Change	0.007** (0.003)	0.007** (0.003)	0.008** (0.003)	0.007** (0.003)
Uploads Per Month	−0.102** (0.043)	−0.106** (0.043)	−0.105** (0.044)	−0.102** (0.043)
Average of Daily Views	0.0001*** (0.00002)	0.0001*** (0.00002)	0.0001*** (0.00002)	0.0001*** (0.00002)
Videos in Cluster 3 (percent)	−0.181 (0.327)			
Videos in Cluster 1 (percent)		−0.537** (0.216)		
Videos in Cluster 2 (percent)			0.537*** (0.196)	0.546*** (0.197)
Average Time to 100 views			−0.025*** (0.005)	
Average Time to 640 views				−0.025*** (0.004)
Average Time to 1000 views	−0.025*** (0.004)	−0.025*** (0.004)		
Constant	−2.766*** (0.131)	−2.559*** (0.152)	−3.319*** (0.155)	−3.120*** (0.160)
Observations	5,297	5,297	5,297	5,297
Log Likelihood	−636.510	−633.469	−638.196	−632.612
Akaike Inf. Crit.	1,285.019	1,278.937	1,288.392	1,277.225

Note: *p<0.1; **p<0.05; ***p<0.01

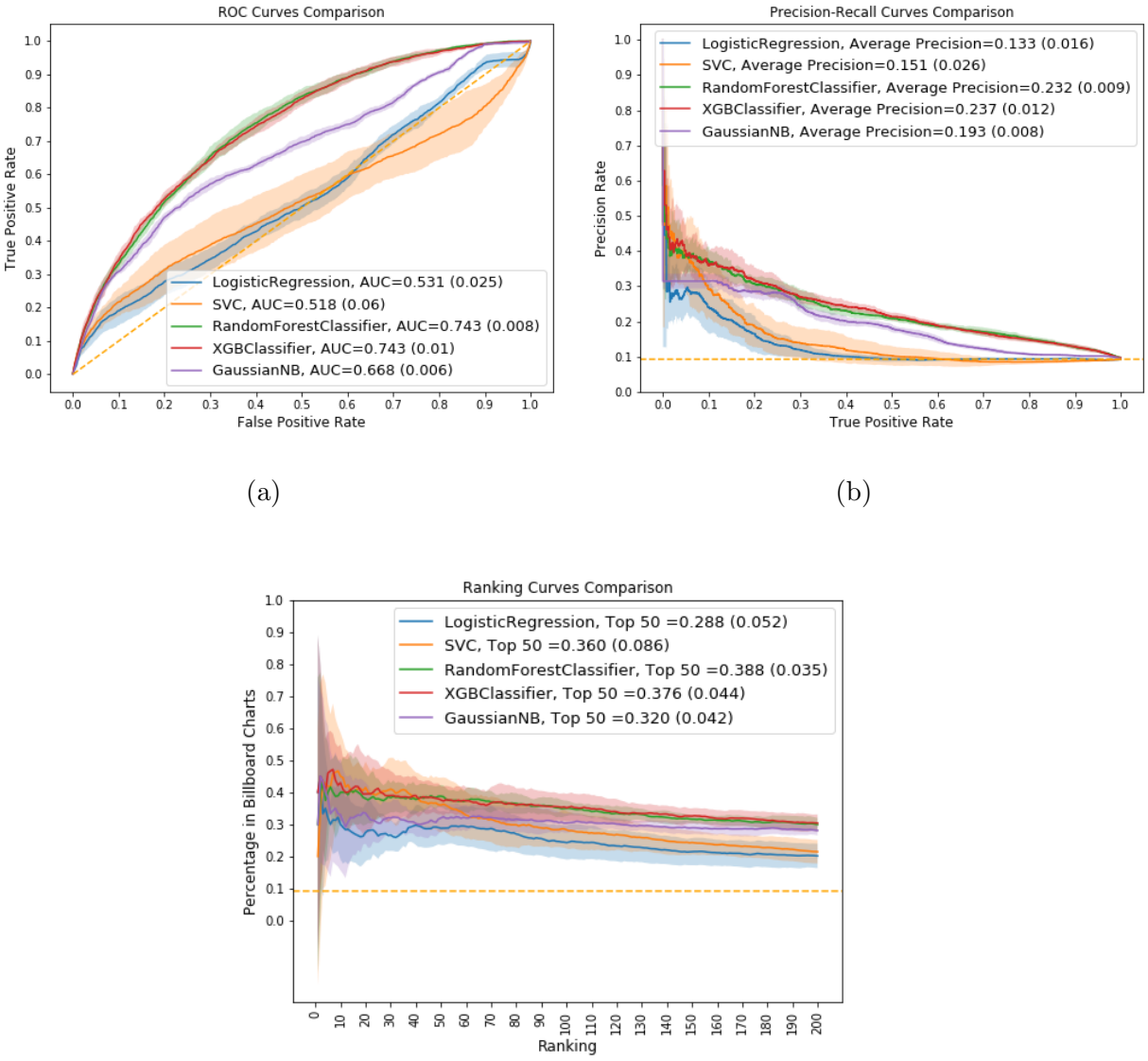
Table A6 Predicting $Signed_i$ using SMOTE to address class imbalance: 10-Run average results for predicting $Singed_i$, whether an artist will work with a major record label in the next period. We report the average and standard deviation for each metric.

classifiers	AUC	Avg. Precision	Precision	Recall	FPR	F1	Accuracy
LogisticRegression	0.614 (0.049)	0.056 (0.009)	0.092 (0.014)	0.244 (0.06)	0.044 (0.009)	0.132 (0.021)	0.943 (0.007)
SVC	0.782 (0.011)	0.087 (0.026)	0.123 (0.015)	0.289 (0.042)	0.038 (0.004)	0.173 (0.021)	0.95 (0.005)
RandomForestClassifier	0.784 (0.013)	0.094 (0.011)	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)	0.982 (0.001)
XGBClassifier	0.744 (0.02)	0.055 (0.005)	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)	0.982 (0.001)
GaussianNB	0.595 (0.048)	0.055 (0.014)	0.055 (0.036)	0.463 (0.241)	0.393 (0.411)	0.086 (0.051)	0.604 (0.399)

Table A7 Predicting Billboard Appearance (anytime): 10-Run average results for predicting $Chart_i$, whether an artist will appear in a Billboard chart in the next period. We report the average and standard deviation for each metric.

classifiers	AUC	Avg. Precision	Precision	Recall	FPR	F1	Accuracy
LogisticRegression	0.531 (0.025)	0.133 (0.016)	0.246 (0.056)	0.105 (0.037)	0.034 (0.014)	0.143 (0.037)	0.887 (0.01)
SVC	0.518 (0.06)	0.151 (0.026)	0.287 (0.127)	0.259 (0.312)	0.213 (0.378)	0.153 (0.026)	0.738 (0.315)
RandomForestClassifier	0.743 (0.008)	0.232 (0.009)	0.368 (0.034)	0.11 (0.026)	0.019 (0.005)	0.168 (0.033)	0.901 (0.003)
XGBClassifier	0.743 (0.01)	0.237 (0.012)	0.371 (0.057)	0.111 (0.034)	0.02 (0.008)	0.167 (0.041)	0.9 (0.006)
GaussianNB	0.668 (0.006)	0.193 (0.008)	0.292 (0.024)	0.16 (0.034)	0.04 (0.012)	0.203 (0.025)	0.886 (0.008)

Figure A2 Predicting Billboard Appearance (anytime): 10-Run Average of Precision-Recall curve (a), the ROC curve (b) and the Accuracy-Ranking curve (c) for predicting $Chart_i$. Shaded regions shows 1 standard deviation from the mean curve. The dotted yellow line in each plot shows the result for a random guess.



(a) (b)

(c)

Table A8 **Results from the Permutation Feature Importance Analysis for predicting $Signed_i$ for Logistic Regression**

Feature Group	Mean Feature Importance	Std. Dev.	Baseline Score
Time to K Views	0.004	0.004	0.021
Average of Video Metrics	0.003	0.003	0.021
Video Metrics Percentage Change	0.000	0.002	0.021
Genres	0.000	0.000	0.021
Video Cluster Percentage	0.000	0.000	0.021
Uploads per Month	-0.000	0.000	0.021
Min-Max Views	-0.003	0.002	0.021

Table A9 **Results from the Permutation Feature Importance Analysis for predicting $Signed_i$ for SVC**

Feature Group	Mean Feature Importance	Std. Dev.	Baseline Score
Min-Max Views	0.044	0.009	0.065
Average of Video Metrics	0.035	0.007	0.065
Uploads per Month	0.000	0.000	0.065
Genres	0.000	0.000	0.065
Video Cluster Percentage	0.000	0.000	0.065
Time to K Views	-0.000	0.000	0.065
Video Metrics Percentage Change	-0.001	0.001	0.065

Table A10 **Results from the Permutation Feature Importance Analysis for predicting $Signed_i$ for Random Forest**

Feature Group	Mean Feature Importance	Std. Dev.	Baseline Score
Average of Video Metrics	0.071	0.007	0.101
Genres	0.016	0.003	0.101
Min-Max Views	0.004	0.002	0.101
Time to K Views	0.003	0.002	0.101
Video Metrics Percentage Change	0.001	0.003	0.101
Video Cluster Percentage	0.001	0.003	0.101
Uploads per Month	-0.001	0.001	0.101

Table A11 **Results from the Permutation Feature Importance Analysis for predicting $Signed_i$ for Gaussian NB**

Feature Group	Mean Feature Importance	Std. Dev.	Baseline Score
Average of Video Metrics	0.033	0.002	0.053
Min-Max Views	0.006	0.004	0.053
Time to K Views	0.003	0.002	0.053
Uploads per Month	0.000	0.000	0.053
Video Cluster Percentage	0.000	0.000	0.053
Genres	-0.000	0.000	0.053
Video Metrics Percentage Change	-0.017	0.003	0.053

Table A12 Results from the Permutation Feature Importance Analysis for predicting $Chart_i$ for Logistic Regression

Feature Group	Mean Feature Importance	Std. Dev.	Baseline Score
Average of Video Metrics	0.012	0.004	0.042
Time to K Views	0.003	0.003	0.042
Video Metrics Percentage Change	0.002	0.001	0.042
Uploads per Month	0.000	0.000	0.042
Genres	-0.000	0.000	0.042
Video Cluster Percentage	-0.000	0.000	0.042
Min-Max Views	-0.029	0.005	0.042

Table A13 Results from the Permutation Feature Importance Analysis for predicting $Chart_i$ for SVC

Feature Group	Mean Feature Importance	Std. Dev.	Baseline Score
Min-Max Views	0.021	0.005	0.058
Average of Video Metrics	0.004	0.002	0.058
Genres	-0.000	0.000	0.058
Video Cluster Percentage	-0.000	0.000	0.058
Uploads per Month	-0.000	0.000	0.058
Time to K Views	-0.000	0.000	0.058
Video Metrics Percentage Change	-0.001	0.002	0.058

Table A14 Results from the Permutation Feature Importance Analysis for predicting $Chart_i$ for Random Forest

Feature Group	Mean Feature Importance	Std. Dev.	Baseline Score
Average of Video Metrics	0.072	0.008	0.116
Video Metrics Percentage Change	0.016	0.005	0.116
Genres	0.002	0.002	0.116
Min-Max Views	0.002	0.002	0.116
Video Cluster Percentage	0.002	0.007	0.116
Uploads per Month	0.000	0.002	0.116
Time to K Views	-0.004	0.004	0.116

Table A15 Results from the Permutation Feature Importance Analysis for predicting $Chart_i$ for Gaussian NB

Feature Group	Mean Feature Importance	Std. Dev.	Baseline Score
Average of Video Metrics	0.041	0.004	0.088
Min-Max Views	0.013	0.004	0.088
Video Metrics Percentage Change	0.011	0.005	0.088
Time to K Views	0.000	0.001	0.088
Uploads per Month	0.000	0.000	0.088
Video Cluster Percentage	0.000	0.000	0.088
Genres	0.000	0.000	0.088

Table A16: **List of Billboard Charts used to create** $Chart_i$**.**

Chart Name	First Date Available	Third column
billboard_adult-contemporary	2005-01-08	2021-04-10
billboard_adult-pop-songs	2005-01-08	2021-04-10
billboard_alternative-airplay	2005-01-08	2021-04-10
billboard_alternative-albums	2007-07-28	2021-04-10
billboard_alternative-digital-song-sales	2011-01-22	2021-04-10
billboard_alternative-streaming-songs	2020-06-13	2021-04-10
billboard_americana-folk-albums	2009-12-05	2021-04-10
billboard_artist-100	2014-07-19	2021-04-10
billboard_australia-digital-song-sales	2009-05-30	2021-04-10
billboard_australian-albums	2011-01-29	2021-04-17
billboard_billboard-200	2005-01-08	2020-05-23
billboard_billboard-argentina-hot-100	2018-11-10	2021-04-03
billboard_billboard-global-200	2020-09-19	2021-04-10
billboard_billboard-global-excl-us	2020-09-19	2021-04-10
billboard_billboard-korea-100	2019-11-02	2021-04-10
billboard_bluegrass-albums	2005-01-08	2021-04-10
billboard_blues-albums	2005-01-08	2021-04-17
billboard_bubbling-under-hot-100-singles	2005-01-08	2021-04-17
billboard_canada-ac	2006-07-22	2021-04-17
billboard_canada-all-format-airplay	2006-08-05	2021-04-17
billboard_canada-chr-top-40	2006-07-22	2021-04-17
billboard_canada-country	2006-07-22	2021-04-17
billboard_canada-emerging-artists	2007-11-03	2021-04-17
billboard_canada-hot-ac	2006-07-22	2021-04-17
billboard_canada-rock	2006-07-22	2021-04-17
billboard_canadian-albums	2005-01-08	2021-04-10
billboard_canadian-hot-100	2007-03-31	2021-04-10
billboard_cast-albums	2006-01-14	2021-04-17
billboard_catalog-albums	2005-01-08	2021-04-10
billboard_christian-airplay	2005-01-08	2021-04-10
billboard_christian-albums	2005-01-08	2021-04-10
billboard_christian-digital-song-sales	2010-01-23	2021-04-10
billboard_christian-songs	2005-01-08	2021-04-10
billboard_christian-streaming-songs	2013-12-07	2021-04-10
billboard_classical-albums	2005-01-08	2021-04-10
billboard_classical-crossover-albums	2005-01-08	2021-04-10
billboard_comedy-albums	2005-01-08	2021-04-17
billboard_compilation-albums	2005-01-08	2021-04-17
billboard_contemporary-jazz	2005-01-08	2021-04-10
billboard_country-airplay	2005-01-08	2021-04-10
billboard_country-albums	2005-01-08	2021-04-10
billboard_country-digital-song-sales	2010-01-23	2021-04-10
billboard_country-songs	2005-01-08	2021-04-10
billboard_country-streaming-songs	2013-04-20	2021-04-10
billboard_current-albums	2005-01-08	2021-04-10
billboard_dance-club-play-songs	2005-01-08	2020-03-28
billboard_dance-electronic-albums	2005-01-08	2021-04-10
billboard_dance-electronic-digital-song-sales	2010-01-23	2021-04-10
billboard_dance-electronic-songs	2013-01-26	2021-04-10
billboard_dance-electronic-streaming-songs	2013-04-20	2021-04-10
billboard_digital-song-sales	2005-01-08	2021-04-10
billboard_emerging-artists	2017-09-02	2021-04-10
billboard_euro-digital-song-sales	2007-12-01	2021-04-10
billboard_france-digital-song-sales	2007-09-08	2021-04-10
billboard_german-albums	2011-02-05	2021-04-17
billboard_germany-songs	2009-07-25	2021-04-17
billboard_gospel-airplay	2005-03-19	2021-04-10

Table A16 – continued from previous page

Chart Name	First Date Available	Last Date Available
billboard_gospel-albums	2005-01-08	2021-04-10
billboard_gospel-digital-song-sales	2010-01-23	2021-04-10
billboard_gospel-songs	2005-03-19	2021-04-10
billboard_gospel-streaming-songs	2013-12-07	2021-04-10
billboard_greece-albums	2011-07-23	2021-04-17
billboard_hard-rock-albums	2007-07-28	2021-04-10
billboard_hard-rock-digital-song-sales	2011-01-22	2021-04-10
billboard_hard-rock-streaming-songs	2020-06-13	2021-04-10
billboard_heatseekers-albums	2005-01-08	2021-04-10
billboard_holiday-albums	2005-01-08	2021-01-09
billboard_holiday-season-digital-song-sales	2010-10-16	2021-01-09
billboard_holiday-songs	2005-01-08	2021-01-02
billboard_holiday-streaming-songs	2013-12-14	2021-01-02
billboard_hot-100	2005-01-08	2021-04-10
billboard_hot-adult-r-and-b-airplay	2005-01-08	2021-04-10
billboard_hot-alternative-songs	2020-06-13	2021-04-10
billboard_hot-canada-digital-song-sales	2005-01-08	2021-04-17
billboard_hot-christian-adult-contemporary	2005-01-08	2021-04-10
billboard_hot-dance-airplay	2005-01-08	2021-04-10
billboard_hot-hard-rock-songs	2020-06-13	2021-04-10
billboard_hot-holiday-songs	2011-12-10	2021-01-02
billboard_hot-mainstream-rock-tracks	2005-01-08	2021-04-10
billboard_hot-r-and-b-hip-hop-airplay	2005-01-08	2021-04-10
billboard_hot-rap-tracks	2005-01-08	2021-04-10
billboard_hot-singles-recurrents	2005-01-08	2021-04-17
billboard_independent-albums	2005-01-08	2021-04-10
billboard_italy-albums	2012-10-06	2021-04-17
billboard_italy-digital-song-sales	2007-09-08	2021-04-10
billboard_japan-hot-100	2011-04-09	2021-04-10
billboard_jazz-albums	2005-01-08	2021-04-10
billboard_jazz-songs	2005-10-22	2021-04-10
billboard_kids-albums	2005-01-08	2021-04-17
billboard_latin-airplay	2005-01-08	2021-04-10
billboard_latin-albums	2005-01-08	2021-04-10
billboard_latin-digital-song-sales	2010-01-23	2021-04-10
billboard_latin-pop-airplay	2005-01-08	2021-04-10
billboard_latin-pop-albums	2005-01-08	2021-04-10
billboard_latin-regional-mexican-airplay	2005-01-08	2021-04-10
billboard_latin-rhythm-airplay	2005-08-13	2021-04-10
billboard_latin-rhythm-albums	2005-05-21	2021-04-10
billboard_latin-songs	2005-01-08	2021-04-10
billboard_latin-streaming-songs	2013-04-20	2021-04-10
billboard_latin-tropical-airplay	2005-01-08	2021-04-10
billboard_lyricfind-global	2015-11-07	2021-04-10
billboard_lyricfind-us	2015-11-07	2021-04-10
billboard_mainstream-r-and-b-hip-hop	2005-01-08	2021-04-10
billboard_mexico	2009-01-17	2021-04-17
billboard_mexico-espanol	2009-01-17	2021-04-17
billboard_mexico-ingles	2009-01-17	2021-04-17
billboard_mexico-popular	2009-01-17	2021-04-17
billboard_new-age-albums	2005-01-08	2021-04-17
billboard_next-big-sound-25	2010-11-13	2021-04-10
billboard_official-uk-albums	2011-01-29	2021-04-17
billboard_official-uk-songs	2011-01-29	2021-04-17
billboard_pop-songs	2005-01-08	2021-04-10
billboard_r-and-b-albums	2013-01-26	2021-04-10
billboard_r-and-b-digital-song-sales	2012-11-10	2021-04-10

Table A16 – continued from previous page

Chart Name	First Date Available	Last Date Available
billboard_r-and-b-hip-hop-digital-song-sales	2010-01-23	2021-04-10
billboard_r-and-b-hip-hop-streaming-songs	2013-04-20	2021-04-10
billboard_r-and-b-songs	2012-10-20	2021-04-10
billboard_r-and-b-streaming-songs	2013-04-20	2021-04-10
billboard_r-b-hip-hop-albums	2005-01-08	2021-04-10
billboard_r-b-hip-hop-songs	2005-01-08	2021-04-10
billboard_radio-songs	2005-01-08	2021-04-10
billboard_rap-albums	2005-01-08	2021-04-10
billboard_rap-digital-song-sales	2010-01-23	2021-04-10
billboard_rap-song	2005-01-08	2021-04-10
billboard_rap-streaming-songs	2013-04-20	2021-04-10
billboard_reggae-albums	2005-01-08	2021-04-17
billboard_regional-mexican-albums	2005-01-08	2021-04-10
billboard_rhythmic-40	2005-01-08	2021-04-10
billboard_rock-airplay	2009-06-20	2021-04-10
billboard_rock-albums	2006-01-14	2021-04-10
billboard_rock-digital-song-sales	2010-01-23	2021-04-10
billboard_rock-songs	2009-06-20	2021-04-10
billboard_rock-streaming-songs	2013-04-20	2021-04-10
billboard_social-50	2010-12-11	2020-12-26
billboard_soundtracks	2005-01-08	2021-04-10
billboard_spain-digital-song-sales	2007-09-08	2021-04-10
billboard_streaming-songs	2013-01-26	2021-04-10
billboard_summer-songs	2010-07-17	2020-09-12
billboard_switzerland-digital-song-sales	2009-05-30	2021-04-10
billboard_tastemaker-albums	2005-12-03	2021-04-17
billboard_top-album-sales	2005-01-08	2021-04-10
billboard_top-triller-global	2020-07-18	2021-04-10
billboard_top-triller-us	2020-07-18	2021-04-10
billboard_traditional-classic-albums	2005-01-08	2021-04-10
billboard_traditional-jazz-albums	2005-01-08	2021-04-10
billboard_triple-a	2005-01-08	2021-04-10
billboard_tropical-albums	2005-01-08	2021-04-10
billboard_uk-digital-song-sales	2009-05-30	2021-04-10
billboard_vinyl-albums	2011-01-22	2021-04-10
billboard_world-albums	2005-01-08	2021-04-17
billboard_world-digital-song-sales	2010-01-23	2021-04-17